



MIDDLESEX Community College

Tools and Technologies for Tech Writers 2025

Homework Helpers

Notices

This document was prepared as a handout for the Middlesex Community College Tools and Technologies for Technical Writers class, Winter semester 2025.

Prepared by Zoë Lawson, course instructor.

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Homework Helpers

This course is very fast paced. A lot of the instructions are vague. This document is an attempt to flush out some details if you need it.

I reiterate this many times over the course: I am covering too much, too fast.

Every single topic I cover could be a full course unto itself.

The intention is that you learn a bit about how you need to learn whatever new tools come your way in the workplace. Do you need a few tutorials? Do you need to buy a book? Do you need someone to train you, hands on?

The other intention is that you figure out if there are tools you hate, so you know to avoid those when job hunting.

Note: This document will be updated each week with step by step instructions to complete the bare minimum homework requirement.

This document is meant to help you if you get stuck. If you perform the tasks in this document, you will pass. However, if all you do is the bare minimum, you won't get a lot out of the course. Your choice.

In this section:

- [Homework requirements](#) on page 5
For each assignment, I am looking to see if you tried to use the tool.
- [Building Blocks](#) on page 6
There are a handful of tasks you need to learn how to do in any program you work in.
- [Suggestion](#) on page 9
Make a simple set of content that you can keep reusing in different tools.
- [Assumptions](#) on page 11
These instructions assume the following:

Homework requirements

For each assignment, I am looking to see if you tried to use the tool.

It is impossible to master any of these tools in a week.

- I do not expect you to get everything correct.
- I do not expect you to become an expert.
- I do not expect this to be an excellent example of your writing abilities.
- I do not expect the assignment to be coherent or logical. It could all be Lorem Ipsum with some formatting applied.

My intention is that it only takes an hour or two to make a simple file (or more, depending upon the tool) that includes the following:

- At least one heading
- A paragraph or two
- An unordered list
- A numbered list
- A table
- Some inline formatting
- A link to an external file
- An image

This text can be completely nonsensical. I just want to see if you tried (not even succeeded) at using the tool.

[Building Blocks](#) on page 6 explains why I selected this set of items.

[Suggestion](#) on page 9 shows an example set of content.

These are the minimum requirements.

If you choose, you can attempt to use each homework assignment to make a portfolio piece. Pick something to document and write it up in whatever the tool of the week is. One of the easiest things to document might be something about whatever tool you are using. If you ask, I will review your content for writing style and offer suggestions.

Building Blocks

There are a handful of tasks you need to learn how to do in any program you work in.

Every tool, and every implementation of a tool, has a unique way of doing things. For example, many content sets require some way to link to the software application they are documenting. This is so if you click the help button in your application, a specific page in your output appears. Most tools offer a way to do this, however how you do it is unique to the tool. Also, what is specifically required is unique to your work place. You may need to set up a mapping file that links application IDs to help IDs, and then do something to add the help IDs to your source files so the "magic" works in the output. You may just need to ensure that output files are named a certain way or put in a certain folder.

That said, there are a relatively small list of things that you need to be able to do in every tool. If you can do these things, you can use the tool.

In this section:

- [As a Writer](#) on page 7

If all you are concerned with is writing, these are the basic tasks you need to know how to do.

As a Writer

If all you are concerned with is writing, these are the basic tasks you need to know how to do.

Required

This is the bare minimum you have to be able to do in any tool to be a technical writer.

- | | |
|----------------------------|--|
| Add a new paragraph | The English language is divided into chunks of text. These chunks are often paragraphs. You need to know how to make a new paragraph as needed. This is usually as simple as pressing Enter , but could require a bit more work, such as remembering to have a blank line between paragraphs in lightweight markup, or having to wrap the text in a <code><p></code> element in HTML. |
| Format the content | <p>As you author content, you need to add formatting to it to help present the information better. You need to know both what formatting to apply, as well as how to do it.</p> <p>Some of this is subjective, such as determining if the content work better as a series of paragraphs, a bulleted list, or a table.</p> <p>Some of this should follow your company's style guide, such as when to apply inline formatting, and knowing the name of the style for a third level heading.</p> <p>Some of this is knowing how your tool works.</p> <p>When I'm describing adding formatting here, in general, I'm referring to applying the formats, styles, or template in your tool. Designing what your format actually looks like is generally something different.</p> |
| Make text a title | The way we structure content in English, your content needs a title. This could be a chapter title, a book title, a section title, or a topic title. This could be done by applying a style to a paragraph, filling in the correct field in a form, or making sure the line of text has a line of equal signs underneath it. |
| Make a list | A lot of technical content requires lists of information: lists of prerequisites, lists of options, lists of things to do next. In many ways, this is just a specialization of <i>Format the content</i> . However, list items are very common. |
| Make a procedure | Many would state that this is the core of technical writing. Our main job is to tell people how to do things. This is often done via a procedure. This could be as simple as making a numbered list, or following a set of styles prescribed by your style guide, or the complex structure of a DITA task topic. Similar to <i>Make a list</i> , this is a specialization of <i>Format the content</i> , but there are often special tools or techniques for working with procedures. |
| Make a table | Tables are a great way to present certain types of information. They also tend to be very complicated. You need to learn how to insert, format, etc. Again, it's another specialization of <i>Format the content</i> , but they are tricky. If you're working with lightweight markup, it can be the most |

Insert an image

complicated formatting. Most tools have wizards and various tools that take some time to learn.

A picture is worth a thousand words. You will need to include images in your content. Different tools do this in different ways.

In general, there are two ways you need to consider inserting images: inline or separately. An inline image is used often in procedures to assist with instructions such as "Click the **Save** () icon".

For larger images, you need to insert them so that they stand alone. That could be inside of a figure with a caption, or just on a separate line. In rare cases, you may configure text wrapping to go around the image.

Insert a link

Content these days usually needs to refer to other things. This can be links to external web sites or links to other parts of your content. (Internal references are often called *cross-references*.)

Linking can be something you have to control manually, or it might be something that can be autogenerated by your tooling. It can also be something you may have to maintain over time, or maybe the tool helps you keep the links working.

Add inline formatting

Besides formatting giant blocks of text, such as paragraphs, lists, and tables, you often need to apply formatting to specific words or phrases in your giant blocks of text. Common examples include making things you click in your software bold, making variables italic, and making command names use a monospace font.

Structure your content

Most likely you don't have a lot to say about the big structure, such as new help systems, new books, or new output formats. However, within the area where you do have control, you will need to know how to structure your content. By structure, this generally means what makes a new section or chapter, and how section headings are nested. This could be making sure you use the right heading level style or could be structuring a bunch of topics into something that makes a table of contents.

Occasional

You may or may not need to do these tasks. These depend upon your tooling, your implementation, and how structured your work place is.

Make a new thing

Most of the time, as a technical writer you are working with content that is already semi-established. The User's Guide already exists, you're just adding new sections to it. You very rarely make a new thing. Also, there's usually a senior writer or information architect who minimally has extremely strong opinions and potentially corporate guidelines to follow about how new large things (new books, new help systems, new top level sections) can be created.

However, if you are working in a topic based system, you will probably often need to make new topics.

Update the table of contents	You will have to structure your content. Depending upon your tooling, your table of contents could be automatically generated, or something you have update. You may also need to be aware of your heading structures to make sure the table of contents works correctly. For example, some tools can't handle if your headings go from heading level 1 to heading level 3, skipping heading level 2.
Indexing or tagging for search	This may not make an actual index any more, but you probably need to do something to improve search results. This can be marking index terms, adding keywords, or adding tags.
Add context sensitive links	If you have a context sensitive help system, you need to add the markers or metadata or whatever to ensure your software can open the right page in your content. This could be something complicated to identify a help ID, or just making sure a file is named something specific.
Work with reuse	This is entirely tooling dependent, but if your tool allows for reuse, you should probably take advantage of it. Reuse can be many different things. You could have certain words or phrases that need to be inserted a specific way, such as version numbers or product names. You could have a way to share topics or chapters between different books. You could be able to reuse specific paragraphs, or any other defined chunk. You will need to learn both how to do the reuse in the tool you are using, and how your company maintains and organizes reused things.
Work with conditional text	Some tools allow you to mark content so that it only appears when certain conditions are met. This means you can mark content that only appears for specific outputs, such as between HTML or PDF output, or based on product or component.

Suggestion

Make a simple set of content that you can keep reusing in different tools.

Every single one of these homework assignments can be portfolio pieces. You can write complicated instructions on how to do things to show off what a great writer you are.

However, to pass this course, all you need to do is try to make all the basic building blocks in different tools.

When trying to learn the different tools, it might be easier if you just have text you are copying and pasting instead of trying to write something new *and* learn how to format something in yet another tool.

I would very much prefer you didn't cut and paste the following, but this is the minimal amount I'm looking for:

Look a Title

The above is a title. It could be formatted as something specifically labeled a title, or maybe just a heading level one.

This is another new paragraph with something **bold** and *italic*. I now have two separate paragraphs, and have some inline formatting.

- I also need a list.
- This is a second list item.

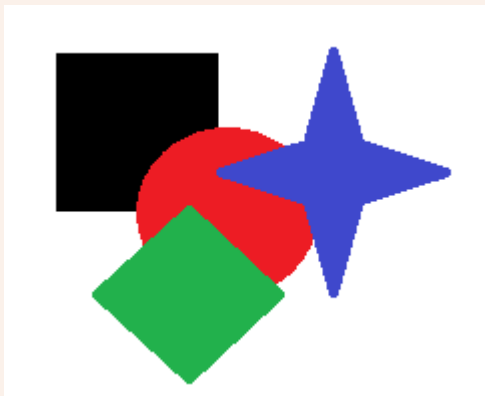
With a paragraph that lines up with the list item.

- And look, a third list item.

I have proven I can make an unordered list. And I can make another paragraph. Oh. I need a link to something, like <http://www.google.com>.

1. Now I need an ordered list.
2. I can use this for procedures.
3. Or to identify items in an image I don't want to translate.

There. Proof I can make a numbered list.



And there is an image. And I remembered to add alternative text.

Table Head 1	Table Head Column 2
Column 1	Column 2
More Column 1	More Column 2
And a third row for fun	With one more column 2

Assumptions

These instructions assume the following:

- You have already cloned the class repository and you are keeping it synced.
- You have Notepad++ installed.

Instructions are in `Handouts/tips_and_tricks.pdf`.

- You have modified Windows File Explorer to always show known file type extensions.

Instructions are in `Handouts/tips_and_tricks.pdf`.

Week 1: Getting Started with Git

The homework for week 1 is to set up your GitHub account.

The homework helper for this week is very minimal because I've tried to have a lot of information in various places.

- `mcc_tools_tech\Week01-IntroGitHub\Week1-IntroGitHub.html` presentation I use during class.

Yes, you are going to have access to all the presentations. While there are versions in existence for the entire class, so you can read ahead and be completely bored during class, but I am updating the presentations before each class, so things will change.

- `mcc_tools_tech\Week01-IntroGitHub\using_git.pdf` (*Using Git*) document that provides more detail than the presentation.

This document is also available in Blackboard.

- `mcc_tools_tech\Handouts\git_cheatsheet.pdf` (*Git Cheatsheet*) document that is more of a reference, but might give a detail in a different way that makes something click.
- <https://drive.google.com/file/d/1ij3FnYD-0f0TFgDDzeVpo74udtSKuTGJ/view?usp=sharing> – A video I threw together in 2022 to help you get started with GitHub.

Everywhere it says "Winter2022", you should use "Winter2025".

Many times in the work place, the training information is slightly out of date. Enjoy learning how to adapt to slightly inaccurate directions.

- [Syncing Repos in GitHub](#) – Another video about syncing repositories in GitHub I created in 2022. You probably don't need it for this week, but you'll definitely need it for next week.

Everywhere it says "Winter2022", you should use "Winter2025".

Week 2: Progressive Information Disclosure

This homework is just being familiar with editing text files.

Be aware that when you're working with computers you often have to deal with *reserved characters*. To make code work, certain words and certain characters mean very specific things.

For example, in our properties file, the equals sign (=) has a specific meaning. Each line of the properties file has the format:

```
Name = Value
```

So, whenever whatever is processing runs into a =, the processor does something special.

For most properties files, the end of a line indicates "done now". So a line break is a reserved character.

Sometimes, you still need to use those special, reserved characters.

To do this, you need to *escape* the special or reserved character. This means you use a different reserved character to say "what follows should be emitted as what it is, don't treat it like the reserved character".

How you escape a reserved character is unique for every language. In general, I just search the internet for "How to escape equals sign in Java properties file" and find out that I generally need to use backslashes (\).

Be aware that the automation I am using is very simplistic. I cannot pass HTML in the properties file and have it work in the generate HTML file.

Week 3 Microsoft Word

Using styles to try and convince Microsoft Word to behave.

The majority of Microsoft Word users are not power users. Most people learned how to write a paper with maybe some running headers and footers and a few cross-references. While you may use it every day in a workplace, you are most likely starting from a template or by editing an existing document.

To produce quality content, you need to make sure you are applying your company's style guide. You need to produce content that is branded appropriately, and follow whatever guidance. You need to use the correct fonts, the right spacing, make sure figures have the correct labeling, etc.

You need to learn to both read the style guide and how to properly apply the styles.

Technically you can be aware of whatever the formatting is, and manually apply it to every paragraph as necessary, but then you're spending more than half your time applying formatting. With styles, you just have to apply the correct style.

In this section:

- [Add Template to Custom Office Templates Folder](#) on page 15
By default Microsoft Word has a specific folder where it expects to find templates. You need to add the template to this folder.
- [Convince Word to find personal templates](#) on page 15
These instructions will be different for every version of Word, but hopefully these get you headed in the right direction.
- [If All Else Fails](#) on page 15
If you are unable to find your Custom Office Templates folder or convince Word to use it, try this method.
- [Things to Remember](#) on page 16
When completing the homework, remember the following.
- [Setup Your Word Document](#) on page 16
When you create a new document using the MCC Word Template, it contains a whole bunch of descriptive text. It explains how to use the template.
- [Add Content to your Word document](#) on page 16
After preparing your Word document, you can start entering your content.
- [Example Word file](#) on page 33
If you followed the previous instructions, your file should resemble the following.

Add Template to Custom Office Templates Folder

By default Microsoft Word has a specific folder where it expects to find templates. You need to add the template to this folder.

You should search the internet for the location of the Custom Office Templates folder for your version of Microsoft Word and your operating system.

Copy `mcc_tools_tech/Week03-WordOffice/Homework/MCC_WordTemplate.dotx` to the Custom Office Templates directory for your system.

If you cannot find this location, you can make your own custom one. Be aware that Microsoft Word can only configure one folder for custom templates. If you change it to a custom directory, you'll have to always put templates here.

Convince Word to find personal templates

These instructions will be different for every version of Word, but hopefully these get you headed in the right direction.

Make sure you have Word templates (.dotx) in your Custom Office Templates, which is usually `C:\Users\User Name\Documents\Custom Office Templates`.

These instructions were determined using Microsoft Office 2021.

1. In Microsoft Word, select **File > Options**.
2. Select **Save** from the navigation pane.
3. In the **Default personal templates location** field, enter the path to where your Word templates are located.
4. Click **OK**.

If all goes well, when you select **File > New**, there should be a **Personal** option.

If All Else Fails

If you are unable to find your Custom Office Templates folder or convince Word to use it, try this method.

1. Double-click the `MCC_WordTemplate.dotx` file.
This should open a new (Document 1) Word document using the template.
2. Use **File > Save As** to save your file.
Technically, you probably just need to use **Save**, but I'd rather be safe than sorry.

Things to Remember

When completing the homework, remember the following.

- Follow the template guidelines.

Unless you are a sole writer at a startup, you are going to walk into a technical writing work place and you are going to have to follow the established styles.

Read the template. It tells you how to use it. (Or at least it tries to.)

- Use the correct styles for the title.
- Use the styles, not the default Word buttons.

Setup Your Word Document

When you create a new document using the MCC Word Template, it contains a whole bunch of descriptive text. It explains how to use the template.

While you could just use **CTRL+A** and then **Delete** to empty out the content and start from scratch, that also means you lose all the helper text, and the information that's set up.

1. Replace the first paragraph ("Company Name") with the company name of the product you are documenting. If your content doesn't have a company, replace the text "Company Name" with Middlesex Community College.

Do not change the style.

2. Replace the text "Product Name" with the name of the product you are documenting. If your content doesn't have a product, replace the text "Product Name" with Tools and Technologies.

Do not change the style.

3. Replace the text "Document Title" with the title of your content.

Do not change the style.

4. Replace the text "Version" with the version of your product. If you don't have one, use the text Week 3.

5. Skip over the table of contents and go to "About this template" on page 3.

6. Select all the text in the rest of the document, starting with "About this Template" and delete it.

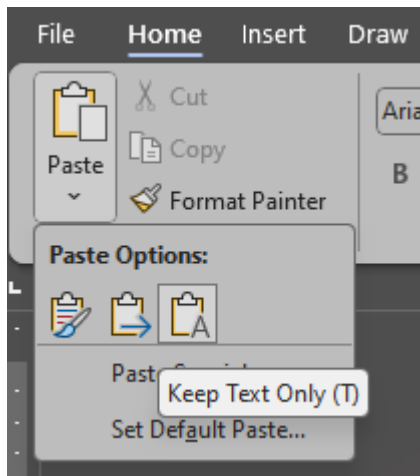
Remember that you don't want to delete the last paragraph. (This may be outdated information, but better to be safe than sorry.)

Add Content to your Word document

After preparing your Word document, you can start entering your content.

You can reformat content you have already written, or you can write new content.

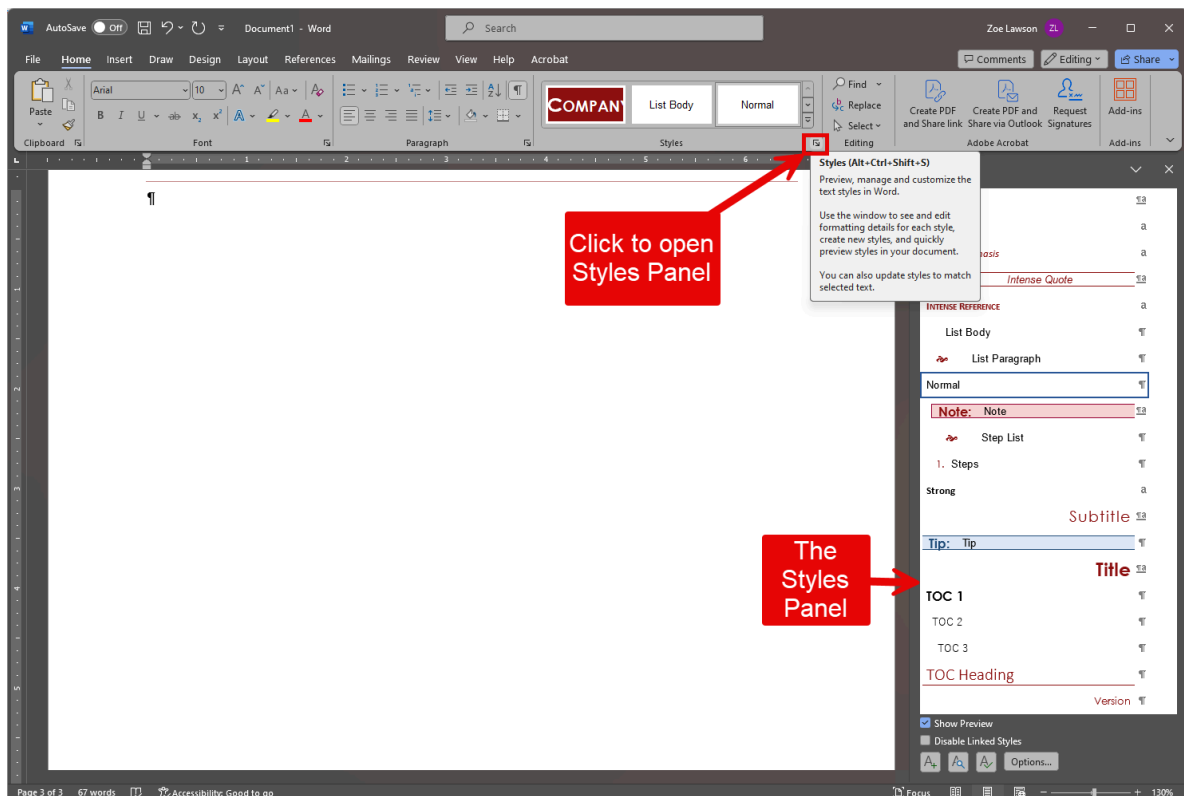
Tip: If you are copying from another Word document, use the Paste option **Keep as Text**.



This makes sure you only get the words, and not extra formatting bits that can easily confuse Word templates.

Make sure you have the Styles panel open.

- On the Home ribbon, click the **Styles** icon to open the Styles panel.



This Styles panel has **Show Preview** enabled so you see a preview of the style, not just the name of the style.

In this section:

- [Add a heading to the Word file on page 19](#)
After setting up your Word document, you should add a level 1 heading.
- [Add a paragraph to the Word file on page 20](#)
After the heading, insert a paragraph.
- [Add inline formatting to the Word file on page 21](#)
Inline or character formatting in Word is done with character styles. Character styles are labeled with a in the Styles panel.
- [Add an unordered list to the Word file on page 23](#)
Unordered lists are sometimes known as bulleted lists.
- [Add an ordered list to the Word file on page 24](#)
Unordered lists are sometimes known as numbered lists.
- [Insert an external link in the Word file on page 25](#)
You often need to link to websites from your documentation.
- [Insert an internal link in the Word file on page 26](#)
You often need to link to other headings in your Word document.
- [Insert an image to the Word file on page 28](#)
Images in Word files are tricky.
- [Add a table to the Word file on page 29](#)
Tables are very useful for displaying data but can be complicated.
- [Update references the Word file on page 30](#)
If you have anything interesting in your Word document, such as a Table of Contents or cross-references, you need to update them from time to time.

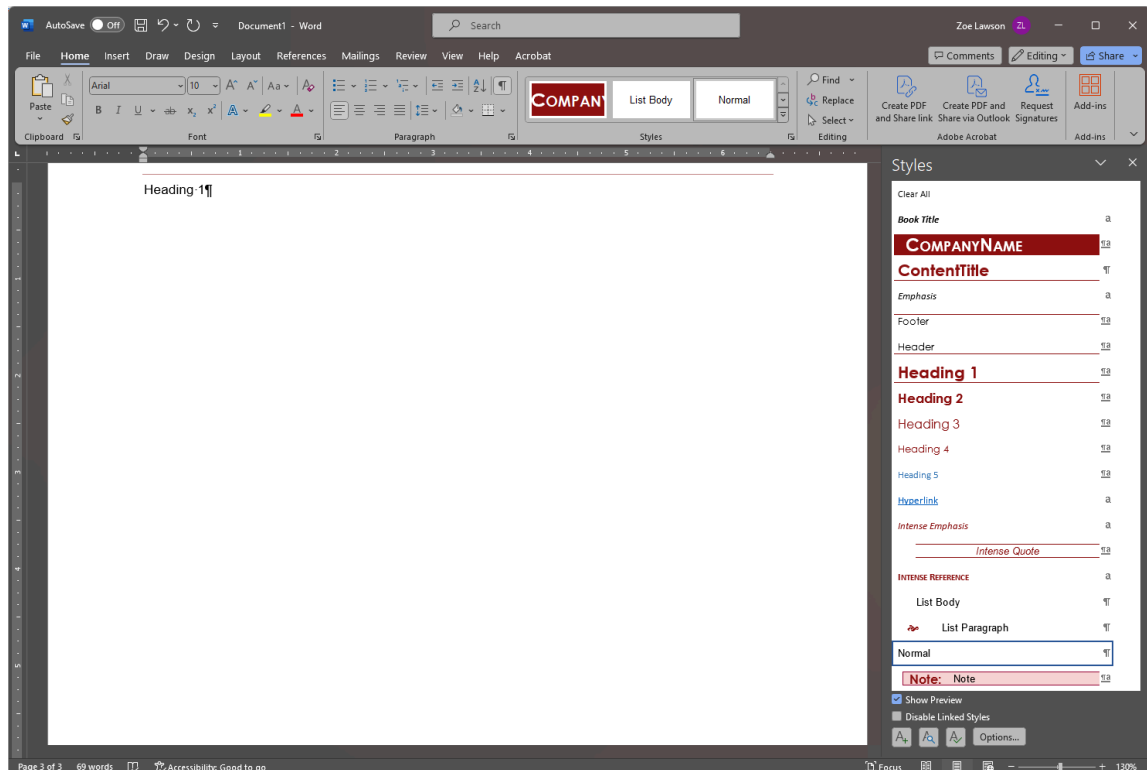
Add a heading to the Word file

After setting up your Word document, you should add a level 1 heading.

You should have a Word document using the MCC Word Template open. Your cursor should be on the first line of page 3 and there should be no content after your cursor.

If your document is not in this state, review [Setup Your Word Document](#) on page 16

1. Type your heading text.

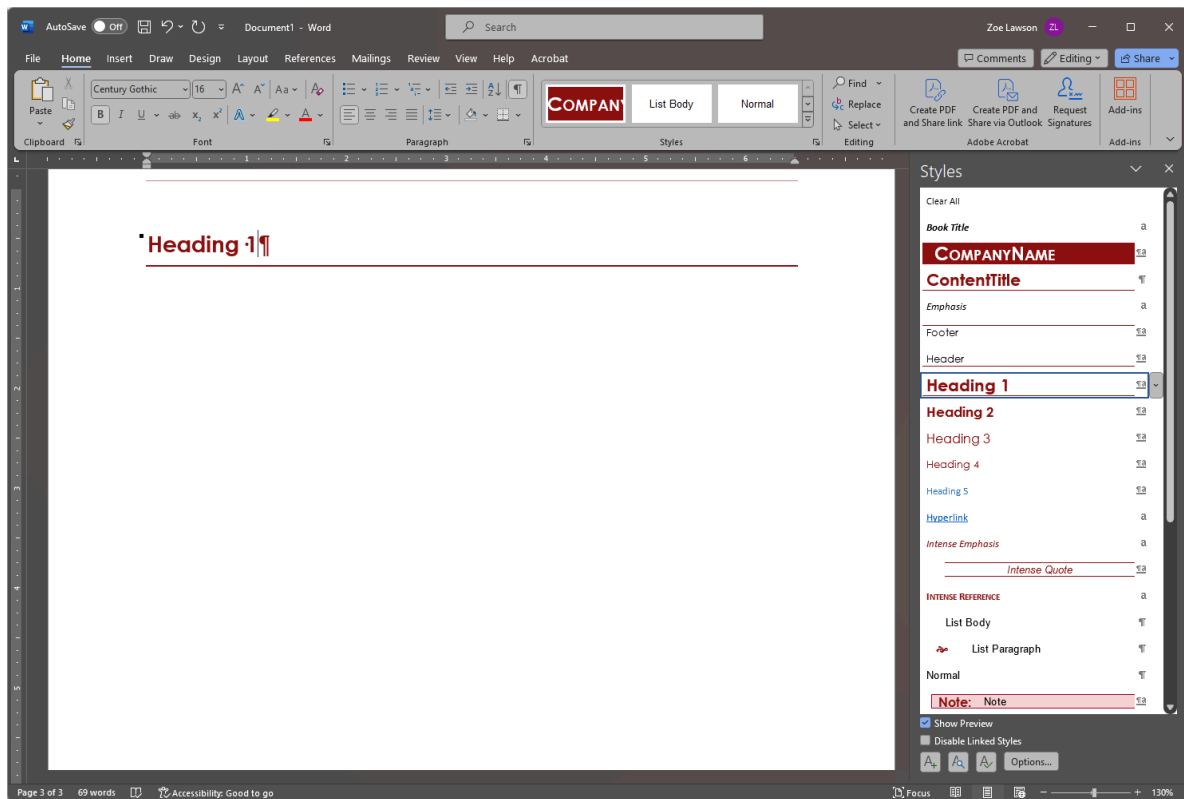


Notice that the style **Normal** is highlighted in the Styles panel.

2. With your cursor still in the paragraph with your Heading 1 text, select the **Heading 1** style.

See that the formatting of the paragraph changes.

Your file should look like the following:



You are now ready to add a paragraph of text.

Add a paragraph to the Word file

After the heading, insert a paragraph.

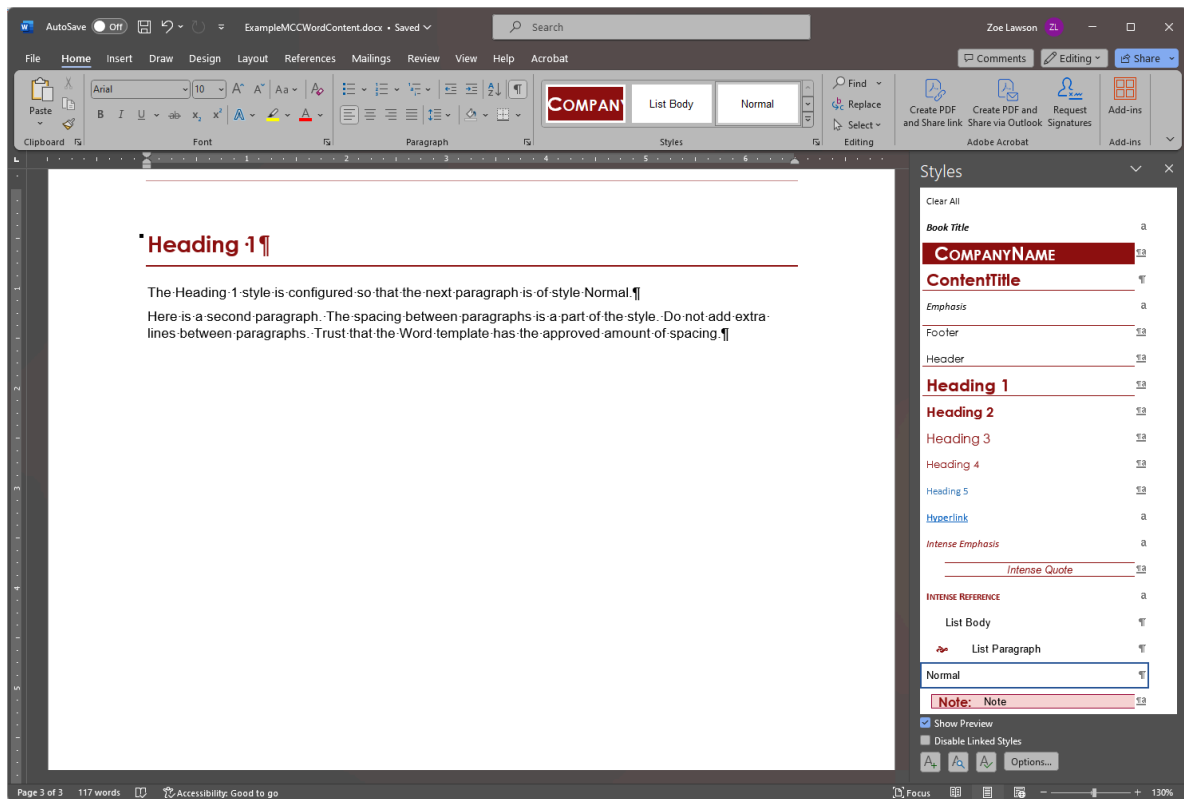
Your cursor should be at the end of your first heading.

1. Press **Enter**.

This creates a new paragraph. The Heading 1 style is configured so that the next paragraph is Normal, the main style for the MCC Word Template.

2. Enter the text of your paragraph.
3. Press **Enter** to add a second paragraph.

You should have a heading with two paragraphs, similar to the following:



Notice the Normal style is highlighted in the Styles panel.

Add inline formatting to the Word file

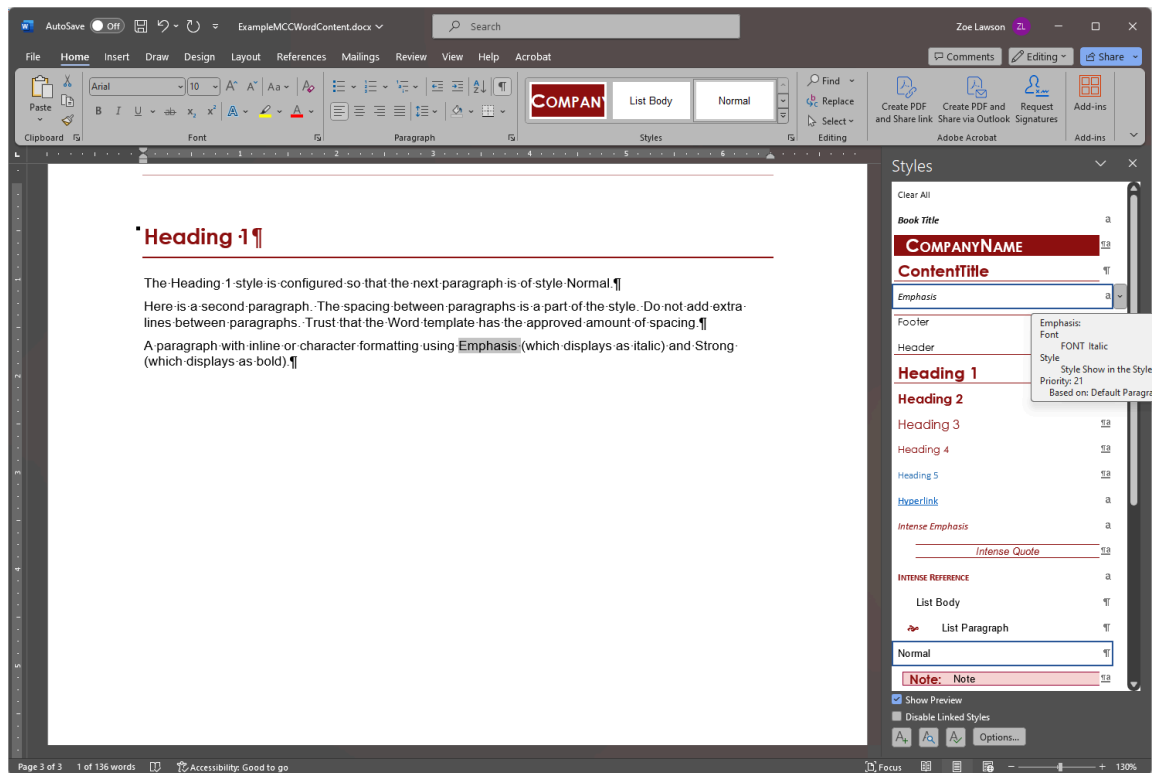
Inline or character formatting in Word is done with character styles. Character styles are labeled with **a** in the Styles panel.

You have a paragraph in your Word file.

1. Make sure you have content in your Word file that you want to change the inline formatting of part of it.

I find it easier to apply character formatting in Word to text that already exists.

2. Highlight the text you want to format and select a Character style.
The following shows applying the Emphasis style.

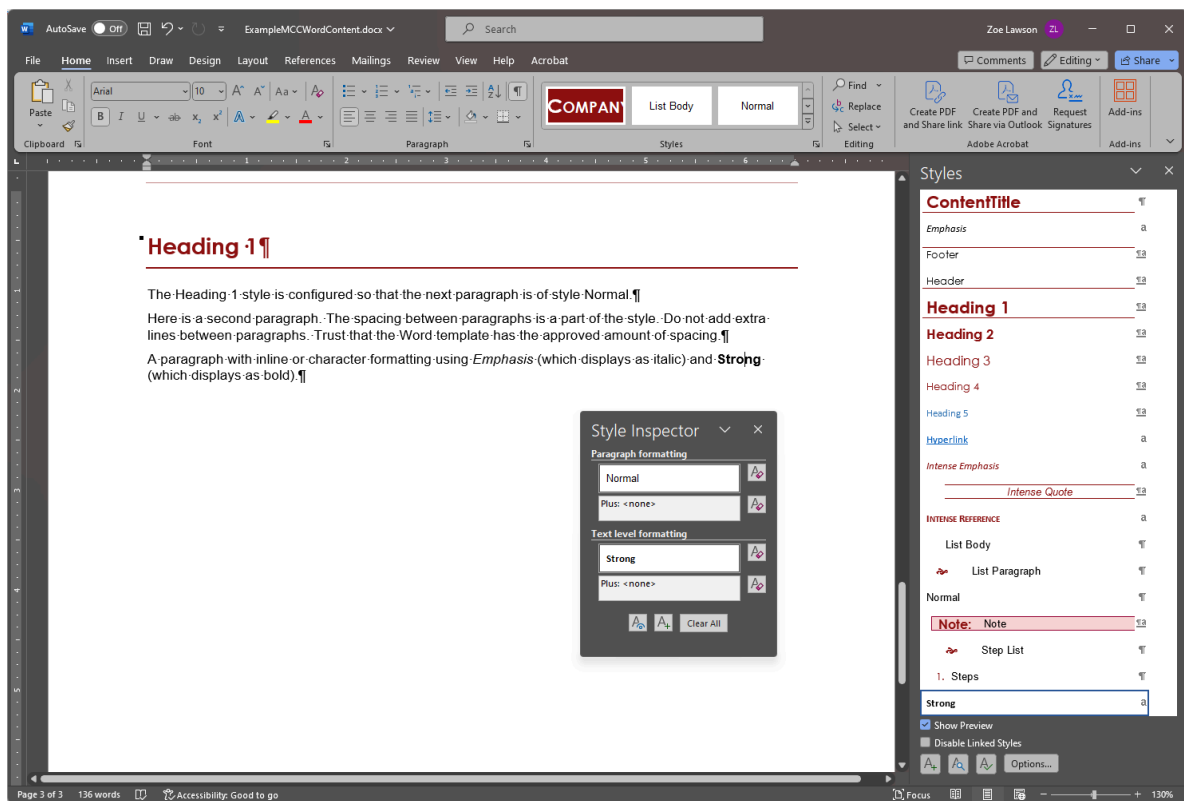


This capture was taken before actually selecting the Emphasis style, so the formatting of the highlighted word hasn't changed yet.

You now have inline formatting marked.

The following shows the Word document with both the Emphasis and Strong character styles applied to content. Also, the cursor is in the text with the Strong character style applied. The Style

Inspector () is open so you can see what styles are applied where your cursor is.



You are ready to make an unordered list.

Add an unordered list to the Word file

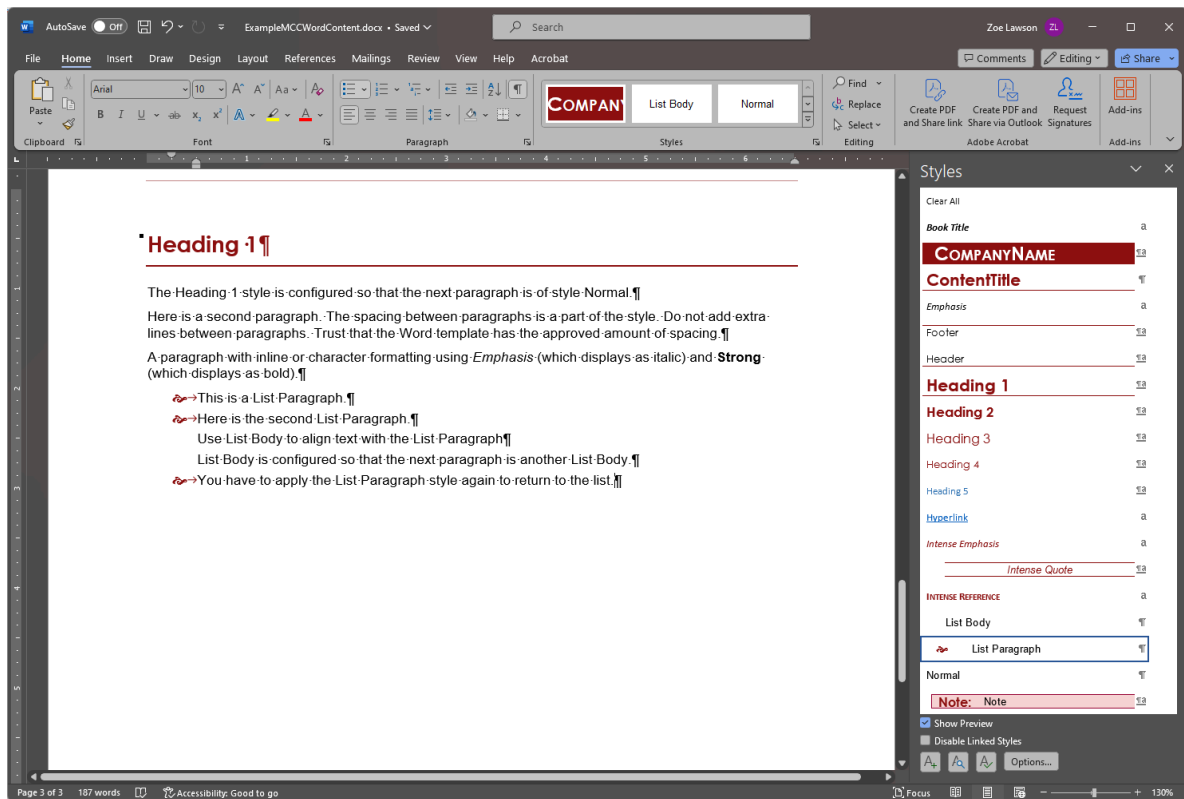
Unordered lists are sometimes known as bulleted lists.

Important: Remember, do not use the Bullets tools on the Home ribbon.

You need to use the appropriate style for the template.

1. Have your cursor at the start of a new paragraph.
2. Select the **List Paragraph** style.
This should indent the paragraph and apply the red squiggle.
3. Press **Enter** to create a second bullet.
List Paragraph is configured so that the next paragraph is another List Paragraph.
4. Use the List Body style to have a paragraph aligned with the text of the List Paragraph.

Your list should resemble the following:



Add an ordered list to the Word file

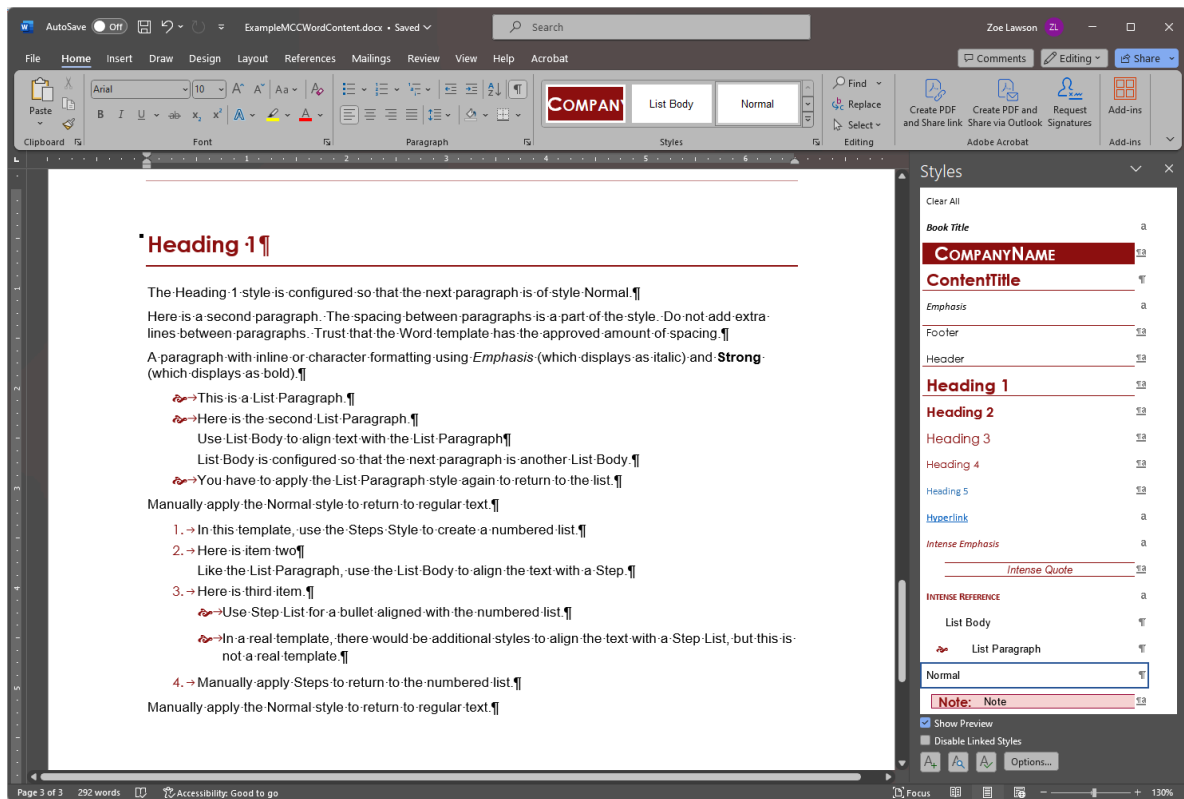
Unordered lists are sometimes known as numbered lists.

Important: Remember, do not use the Bullets tools on the Home ribbon.

You need to use the appropriate style for the template.

1. Have your cursor at the start of a new paragraph.
2. Select the **Steps** style.
This creates a numbered list in the correct style for this template.
3. Use the **List Body** style to create a paragraph that aligns with the step text.
4. Use the **Step List** style to create a bulleted list that aligns with the step text.

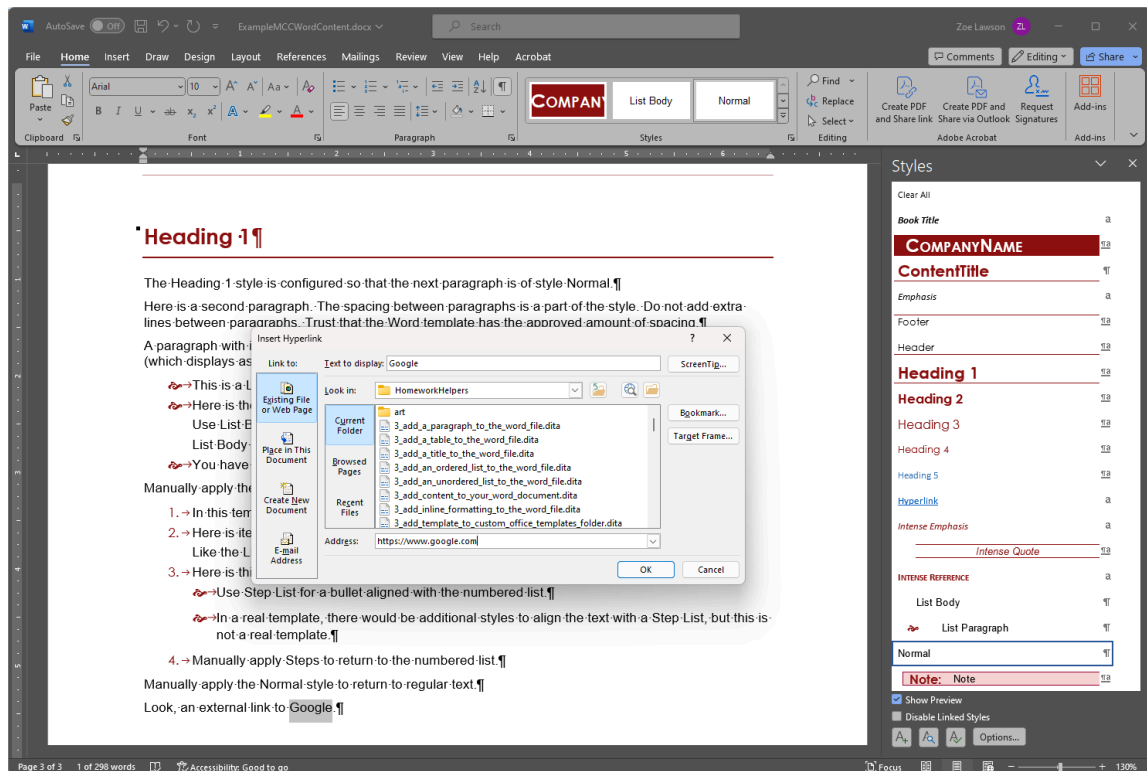
Your list should resemble the following:



Insert an external link in the Word file

You often need to link to websites from your documentation.

1. Have your cursor where you want to insert your link.
2. Type the text for your link.
3. Select the text.
4. Press **CTRL + K** or from the **Insert** ribbon, click **Link**.
The **Insert Hyperlink** dialog box opens. Notice that the **Text to display** field contains the text you selected.
5. In the **Address** field, enter the URL.
In this case, enter <https://www.google.com>.



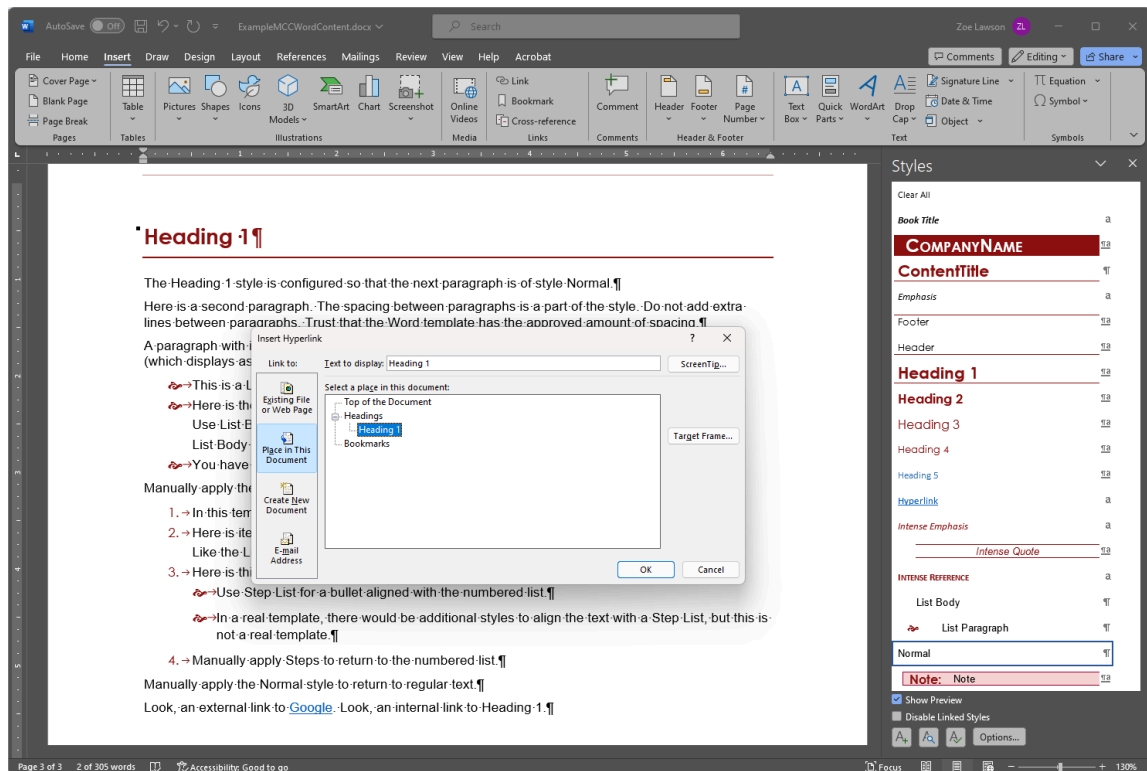
6. Click **OK** to finish entering the link.

Insert an internal link in the Word file

You often need to link to other headings in your Word document.

Word only lets you link to predefined locations. You can link to paragraphs formatted with the official Word heading styles or to bookmarks.

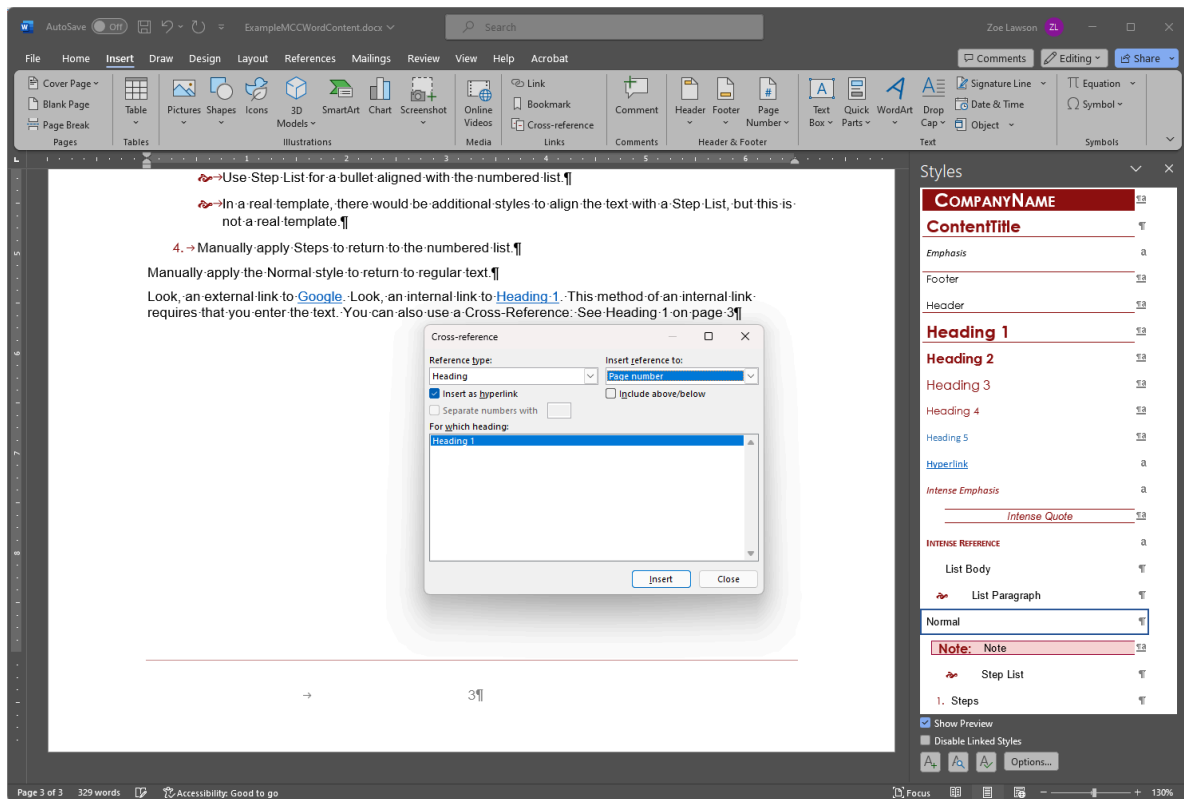
1. Have your cursor where you want to insert your link.
2. Type the text for your link.
3. Select the text.
4. Press **CTRL + K** or from the **Insert** ribbon, click **Link**.
The **Insert Hyperlink** dialog box opens. Notice that the **Text to display** field contains the text you selected.
5. In the **Link to** list, select **Place in This Document**.



6. Select a heading that exists in your document.

7. Click **OK** to finish entering the link.

Alternatively, you can use a cross-reference by selecting **Cross-reference** from the Insert ribbon. Cross-references let you choose between a set of style types, including headings. You can insert the heading text or the page number. You can also combine the two (manually typing on page) to have a "See heading on page" cross-reference. Cross-references should update if you change heading text after you Update References in the document.



Insert an image to the Word file

Images in Word files are tricky.

I believe it is technically possible to insert an image by reference, but generally Word pastes the image into the file.

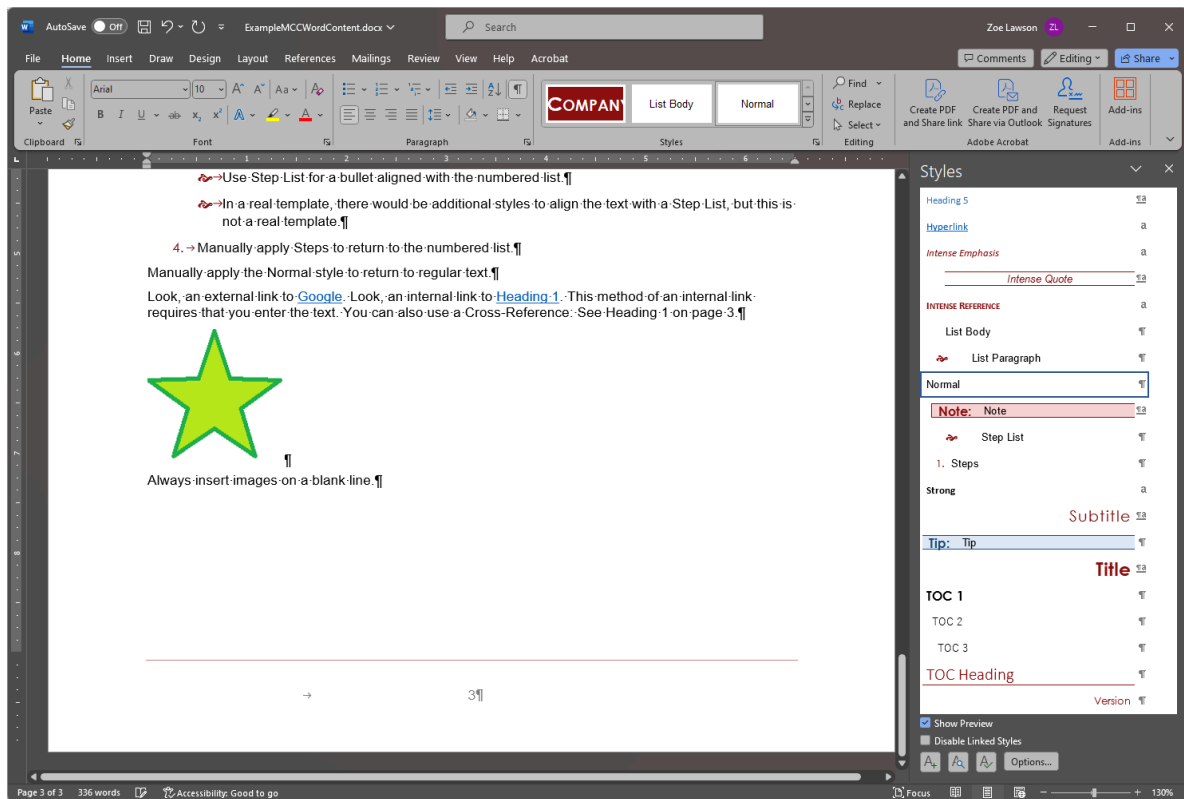
Word does not really support high-resolution images for print.

While Word does have some limited embedded image editing tools, do not use them. Always edit your image in a graphics editor. Don't use the resizing, cropping, or markup tools in Word. When you use the Word graphic tools, you lose image quality, and there is no guarantee how those edits will transfer to other tools. (For example when converting Word files to HTML using third-party tools like WebWorks ePublisher.)

Follow the guidance of your company's template.

I have never been able to get Word figure captions to work the way I expect. May you have better luck.

1. Have your cursor where you want to insert your image.
Whenever possible, insert your image in a blank paragraph. This way you might run into fewer issues with the image jumping pages.
2. From the Insert ribbon select **Pictures > (Insert Picture From) This Device**.
3. Navigate to the image you want to insert and select it.
4. Click **Insert**.

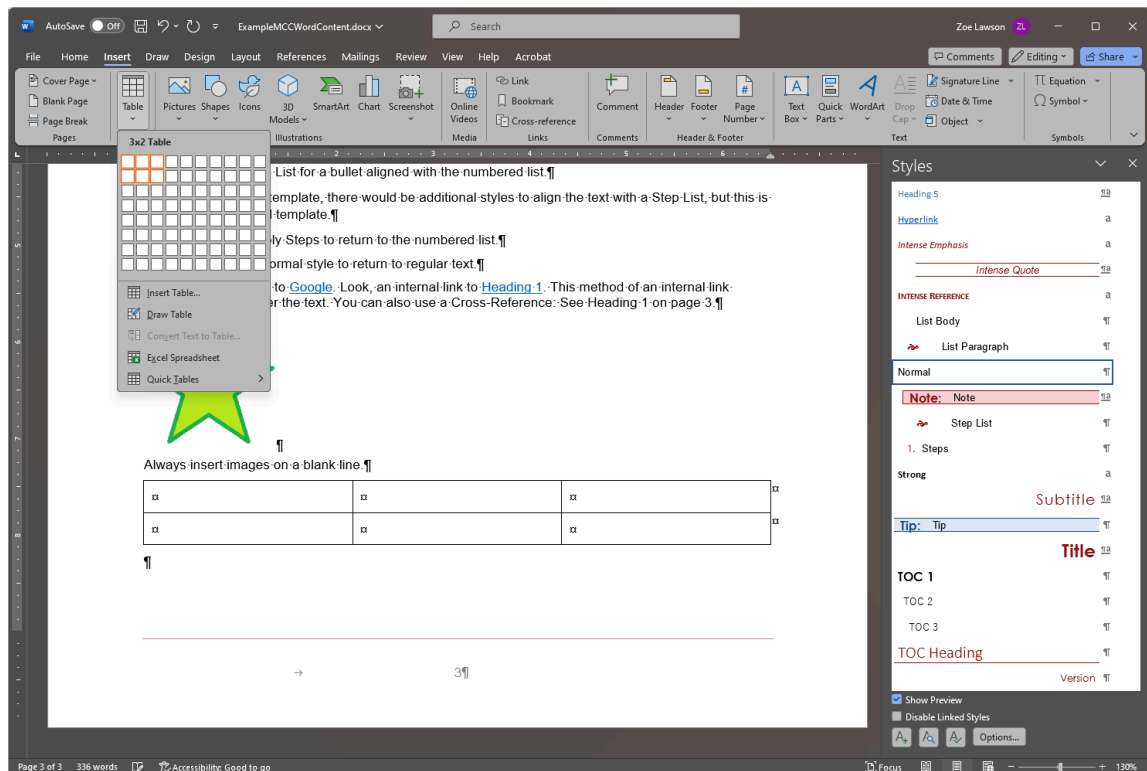


Add a table to the Word file

Tables are very useful for displaying data but can be complicated.

Unfortunately, I have not defined tables in the MCC Template very well. Hopefully if you ever have to work in Word, you will have a better defined template.

1. Similar to images, insert tables on a blank line.
2. From the Insert ribbon, select **Table** and select how many rows and columns you want to insert initially.



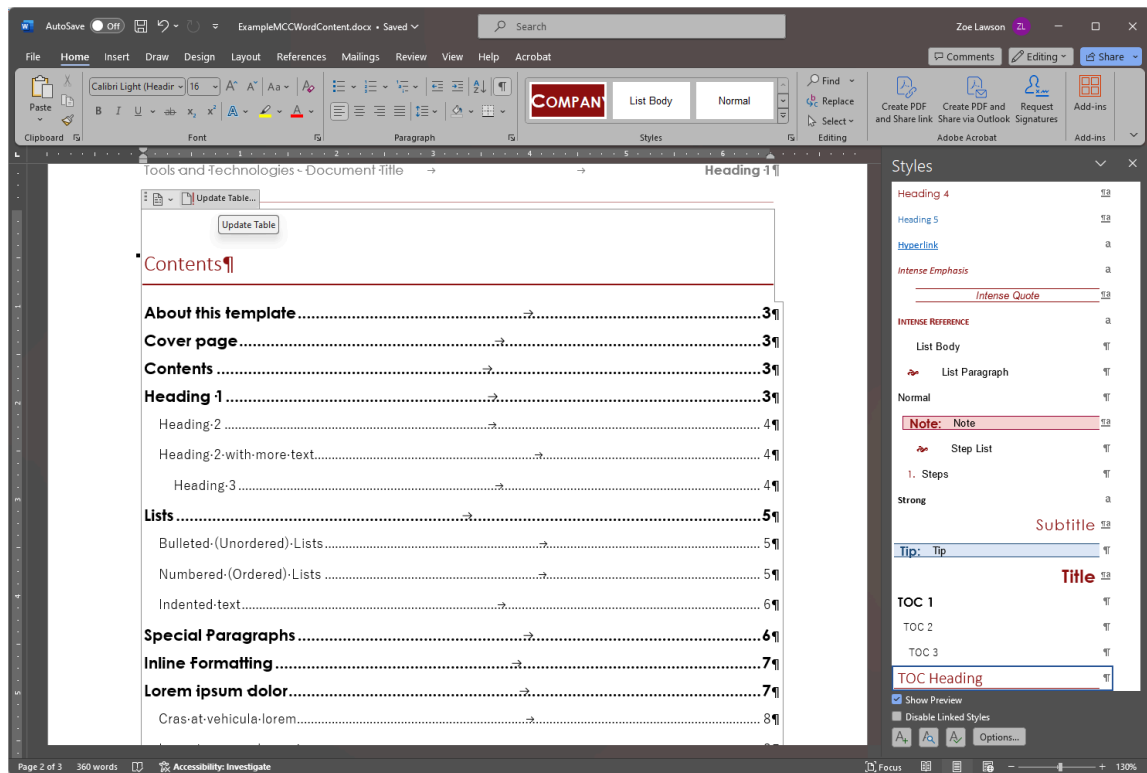
3. If you have a well defined template, there are Table styles from the Table Design ribbon. These styles define heading rows, shading, etc.
4. When you are in a table, continue using Styles. A well defined template will have special styles for table text.

Update references the Word file

If you have anything interesting in your Word document, such as a Table of Contents or cross-references, you need to update them from time to time.

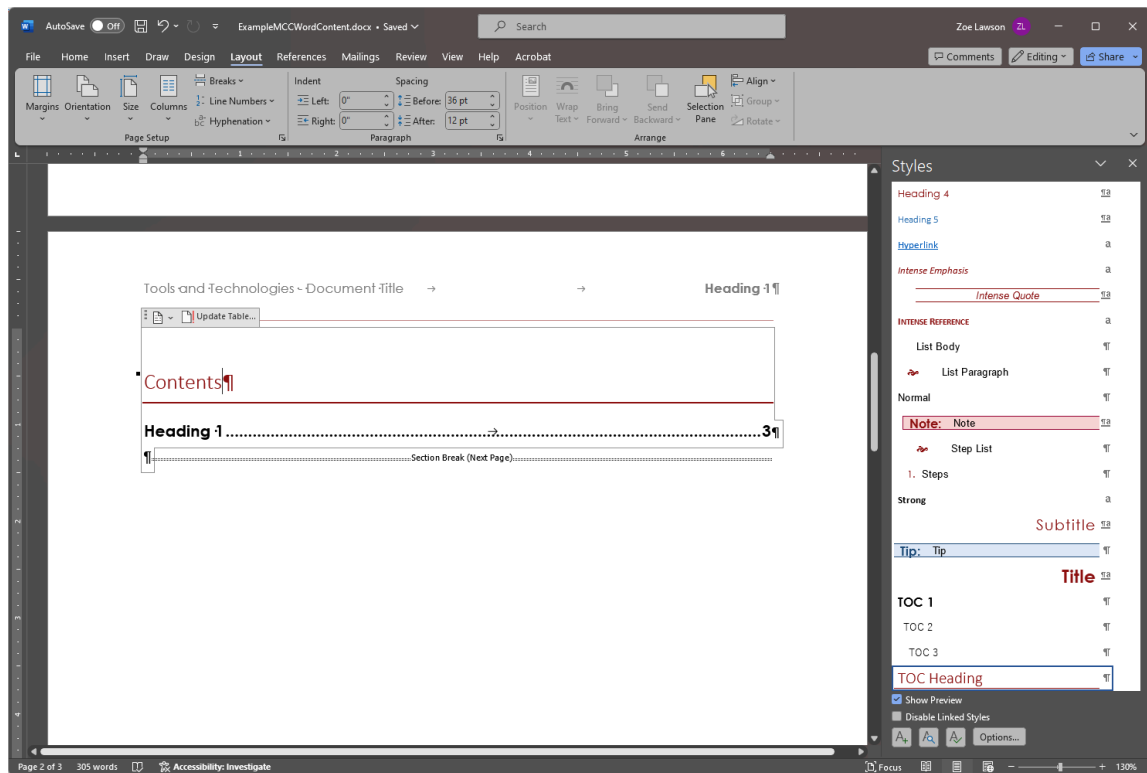
Unfortunately, these commands don't always work as well as they should, because Word.

1. If all you have is a Table of Contents, select the Table of Contents.
At the top of the Table of Contents, some popup controls should appear.



2. Select **Update Table** and the table of contents should regenerate.

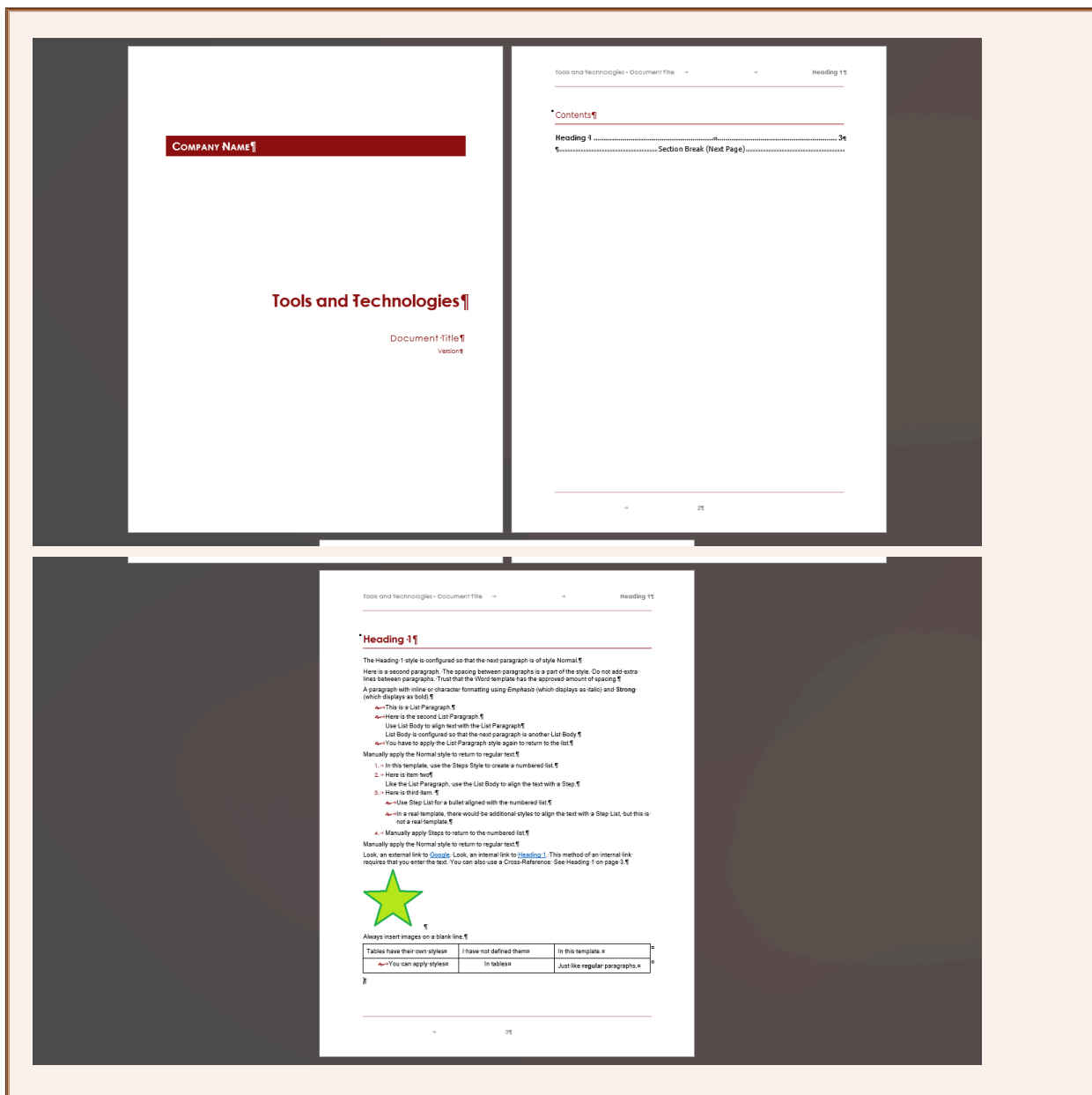
In this example, we only added a single Heading 1, so the table of contents is very simple.



3. If you have other references, such as cross-references:
 - a) Select the entire document using **CTRL + A**.
 - b) Right-click the selected text and choose **Update Fields**.If all goes well, all cross-references and page numbers update.

Example Word file

If you followed the previous instructions, your file should resemble the following.



Save your file and upload it to GitHub. If you need a refresher on working with GitHub, see Handouts/git_cheatsheet.pdf.

Congratulations! You've made a Word document using the MCC Template!

Week 4 Tours

Using PowerPoint to mock up a tour.

Actually creating a tour requires software that needs a tour and developers that help you implement the tour. If you're lucky, you may have an overlay tool such as WalkMe.

In many cases, you will want to plan out your tour before you start coding it. A simple way to do that is using PowerPoint (or whatever your favorite presentation tool is).

For this course, you want to make a very simple tour.

- You only need 4-6 steps.
- Cover something that doesn't require changing the screen.

If you start wanting to change what screen you're looking at, that probably should be a video or tutorial.

The following instructions are for PowerPoint, but you can use any tool you'd like.

In this section:

- [Take a Screen Capture of Your Software](#) on page 34
There are many tools out there, but all operating systems should have a way to take a screen capture..
- [Set up a New PowerPoint Presentation](#) on page 35
You don't have to use PowerPoint, but this tends to be the easiest way.
- [Add a Slide](#) on page 35
After setting up your presentation, you can start adding slides and "tour bubbles".

Take a Screen Capture of Your Software

There are many tools out there, but all operating systems should have a way to take a screen capture..

We discuss this in more detail when we talk about graphics, but there are many tools available for taking screen captures. My current favorite is TechSmith SnagIt. However, here we're just going to discuss built in tools.

- Windows: Use **ALT + Print Screen**.

You can use just **Print Screen**. Print Screen grabs your entire screen, including multiple monitors. Using ALT and Print Screen just grabs the active window.

The image is saved to your clipboard, so you should be able to just paste it where you need it.

Newer versions of Windows may open the Snipping Tool which can provide more options.

- Mac: Use **SHIFT + COMMAND + 3**.

The image is saved to the desktop, according to the internet.

You could also use an app on your phone for making a tour. You can look up how to take a screen capture on your phone.

Set up a New PowerPoint Presentation

You don't have to use PowerPoint, but this tends to be the easiest way.

1. Make a new PowerPoint presentation, using a blank presentation.
2. On the **View** ribbon, select **Slide Master**.
3. Select the Blank layout from the left list of layouts.
4. Paste the screen capture you took onto the Blank layout.
5. Return to the Normal View. On the **Slide Master** ribbon, select **Close Master View**, or on the **View** ribbon, select **Normal**.

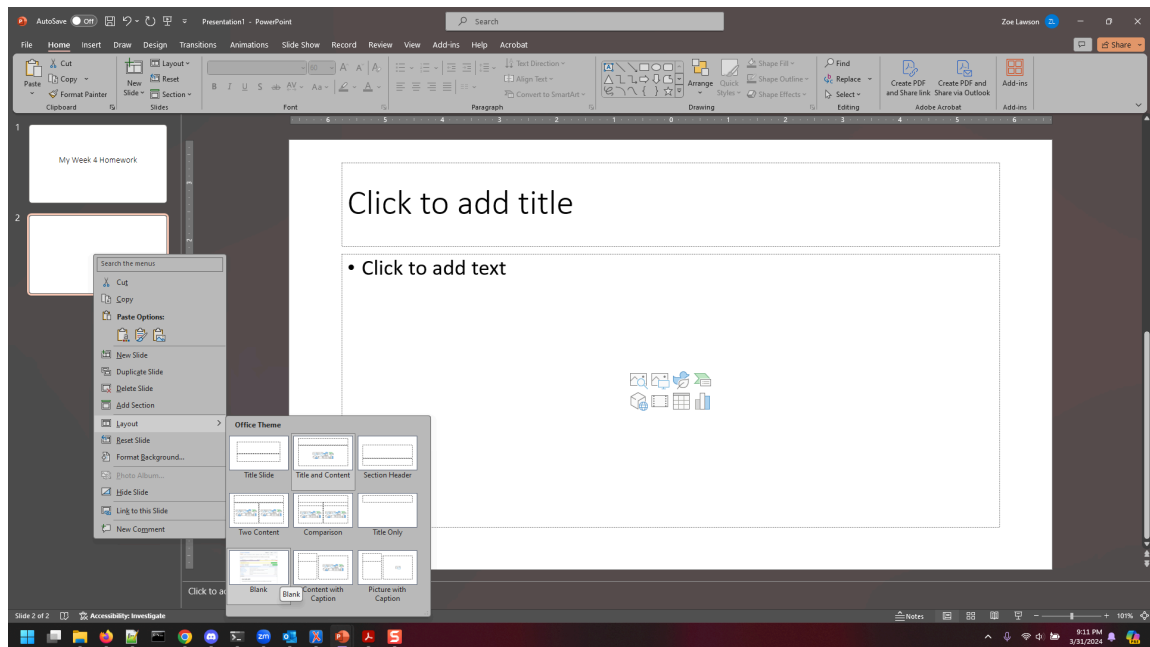
Now the screen you want to make a tour for is available when ever you apply the Blank layout to a slide in the presentation.

Don't forget to save the file. Save early, save often.

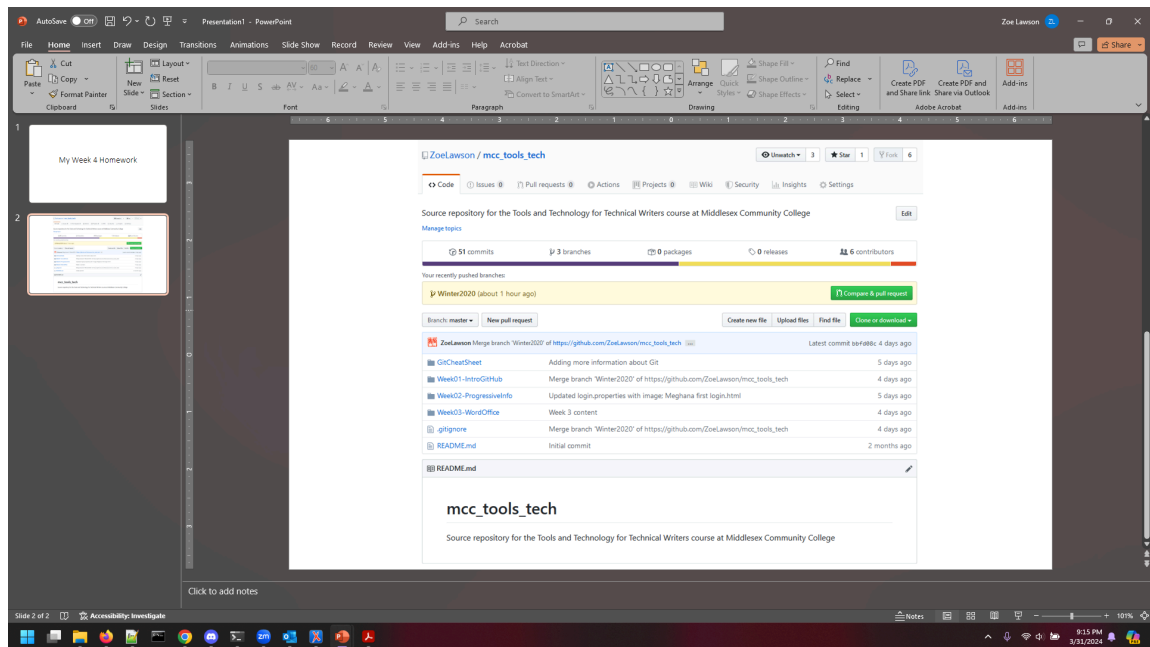
Add a Slide

After setting up your presentation, you can start adding slides and "tour bubbles".

1. In your PowerPoint presentation in Normal view, make a new slide using the Blank layout.
To change the layout of a slide, right-click the slide and select **Layout > Blank**.

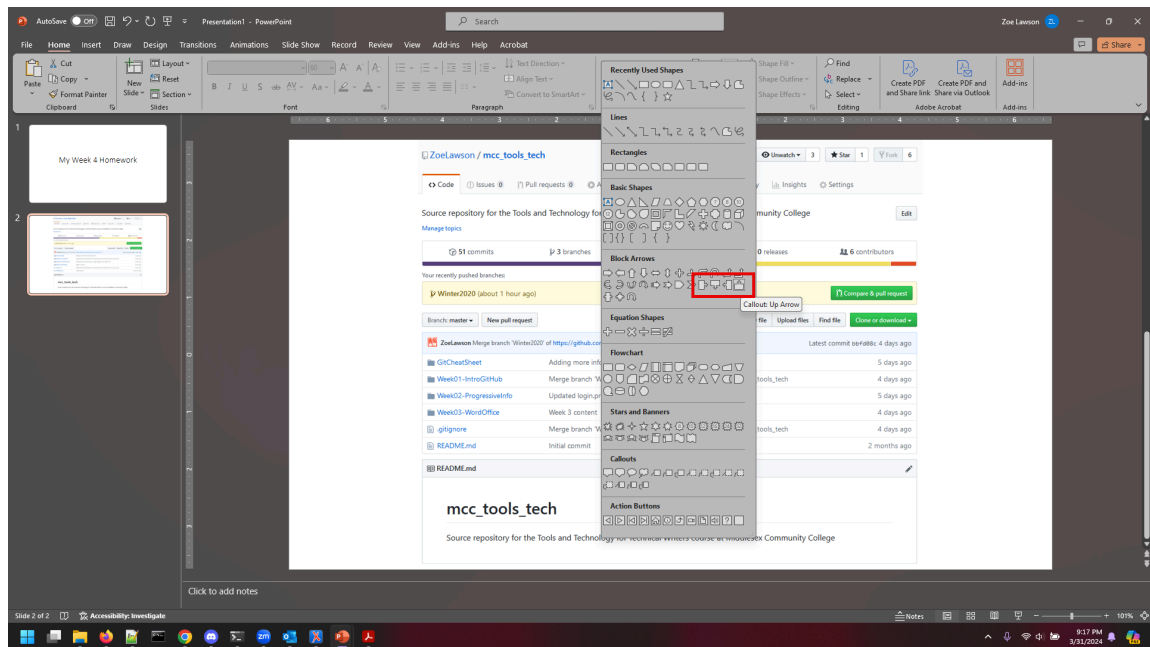


You should get a new slide with your screen capture of your software on it.

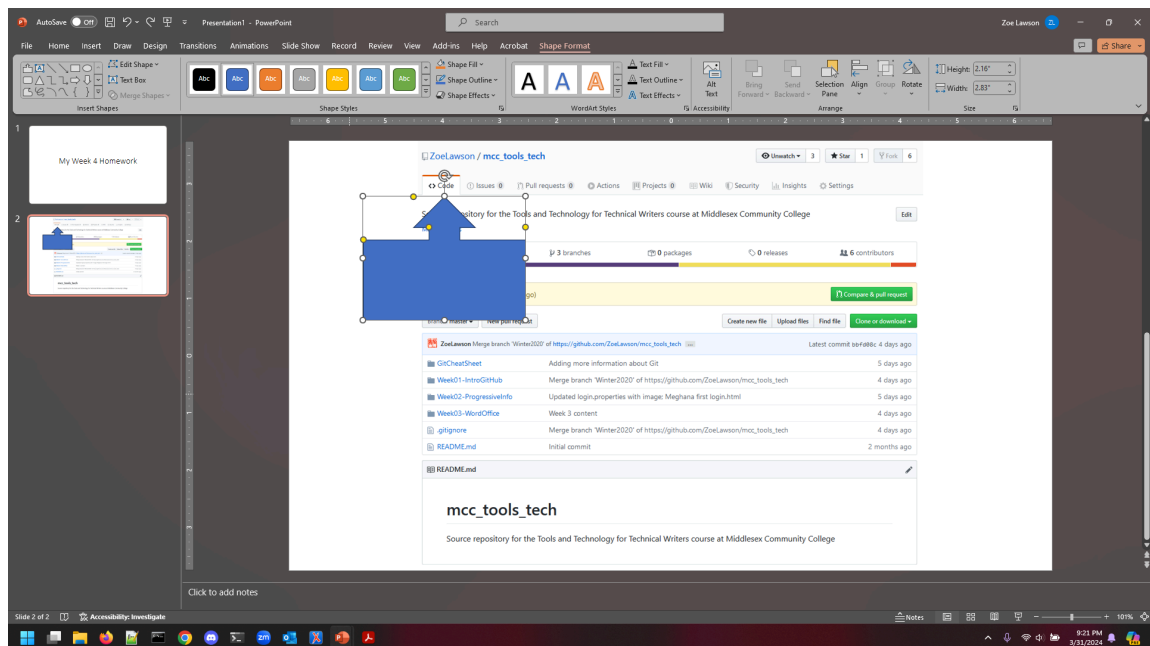


2. Add a "tour bubble" by selecting a shape.

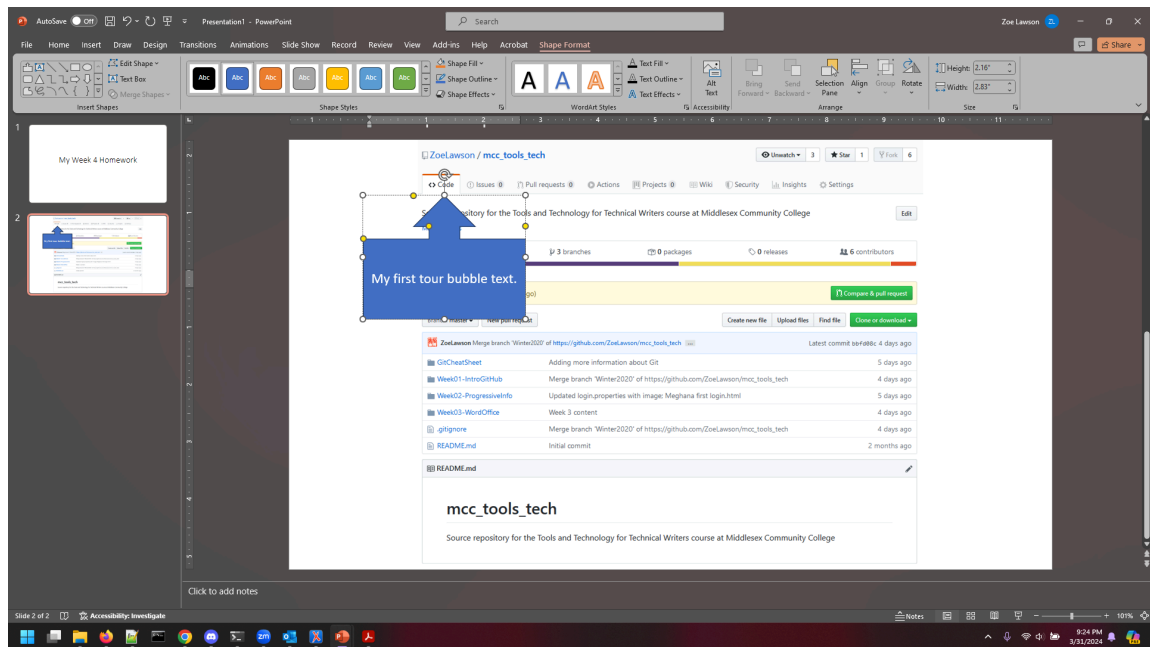
You can use any shape you'd like, but I'd recommend something that includes a pointer to the part of the screen you care about.



After selecting the shape, draw it on the slide by using click and drag. You can resize and position the shape as needed.



3. Double-click in the shape to activate the text mode and add your text.



Repeat for the rest of your slides.

If you need more of an idea about what to do, look at the `mcc_tools_tech\Week04-Tours\Homework\ExampleTour.pptx` file.

Week 5 HTML

A too short introduction to HTML (and CSS).

Since HTML is how text displays in browsers, and that's how 90% of the non-video content on the web is presented, knowing a smattering of HTML is extremely useful.

In general, no one writes HTML pages by hand any more. You author in one tool and export or transform it into HTML. Every now and then, however, you can format text using basic HTML in other tools. This should help you learn the basics.

In this section:

- [Make an HTML file](#) on page 40
HTML is a text file.
- [Open the HTML file in Notepad++](#) on page 41
There are many HTML editors out there, but we're going to use Notepad++.
- [Set up the HTML file](#) on page 41
HTML has some text that's required.
- [Add a title to the HTML file](#) on page 43
You can add content in most any order in an HTML file, but in general, you start with a title or heading.
- [Add a paragraph to the HTML file](#) on page 44
After the title, insert a paragraph.
- [Add inline formatting to the HTML file](#) on page 45
Inline formatting in HTML is done with additional tagging.
- [Add an unordered list to the HTML file](#) on page 46
Unordered lists are sometimes known as bulleted lists.
- [Add an ordered list to the HTML file](#) on page 47
Unordered lists are sometimes known as numbered lists.
- [Insert an external link in the HTML file](#) on page 48
You often need to reference external websites when writing.
- [Insert an internal link in the HTML file](#) on page 49
Internal links, meaning to some paragraph or heading on the same page are also useful.
- [Insert an image in the HTML file](#) on page 50
Images in HTML are inserted by reference.
- [Add a table to the HTML file](#) on page 51
Tables are very useful for displaying data, but can be complicated.
- [Example HTML file](#) on page 54
If you followed the previous instructions, your file should resemble the following.

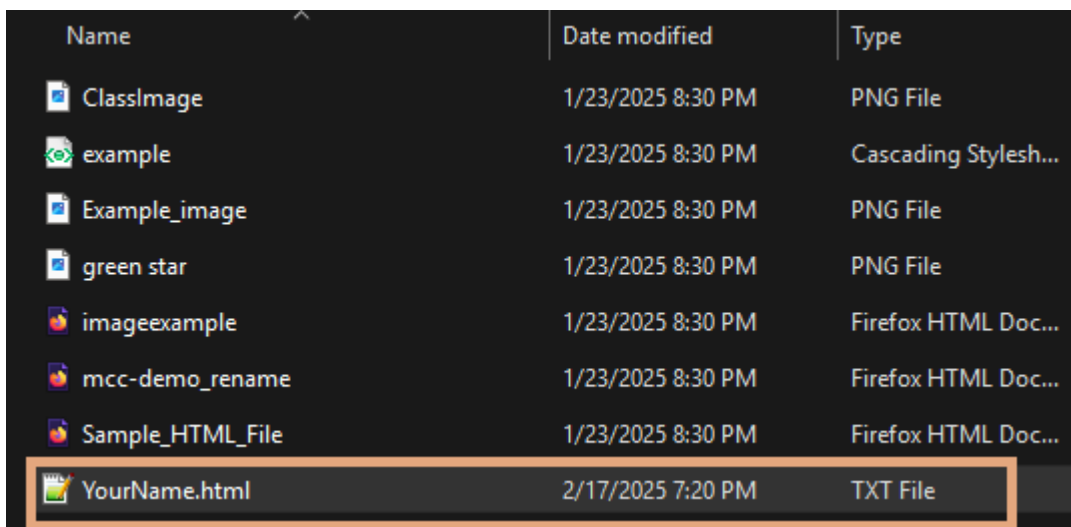
Make an HTML file

HTML is a text file.

1. In File Explorer, right-click in the `mcc_tools_tech\Week05-AgileHTML\Homework` folder and select **New > Folder**.
This makes a new folder in the Homework directory. It should be called `New Folder`, and the text "New Folder" should be highlighted and ready to rename the folder.
2. Rename the folder *YourNameHW*.
For example, I would name the folder `ZoeLawsonHW`.
I will keep referring to the folder as *YourNameHW*.
3. Navigate into the *YourNameHW* folder.
4. Still in File Explorer, right-click in *YourNameHW* and select **New > Text Document**.
This makes a new text document in the *YourNameHW* folder. It should be called `New Text Document.txt`, and the text "New Text Document" should be highlighted and ready to rename the file.
5. Rename the text file *YourName.html*.
For example, I would name the file `ZoeLawson.html`.
You must change the extension to `html`.

You should receive a message stating If you change a file name extension, the file might become unusable. Are you sure you want to change it?.

If you do not receive the above message, you may have your system configured to hide known file extensions. Look at your file in File Explorer. If it looks like the following, it's saved as a text file.



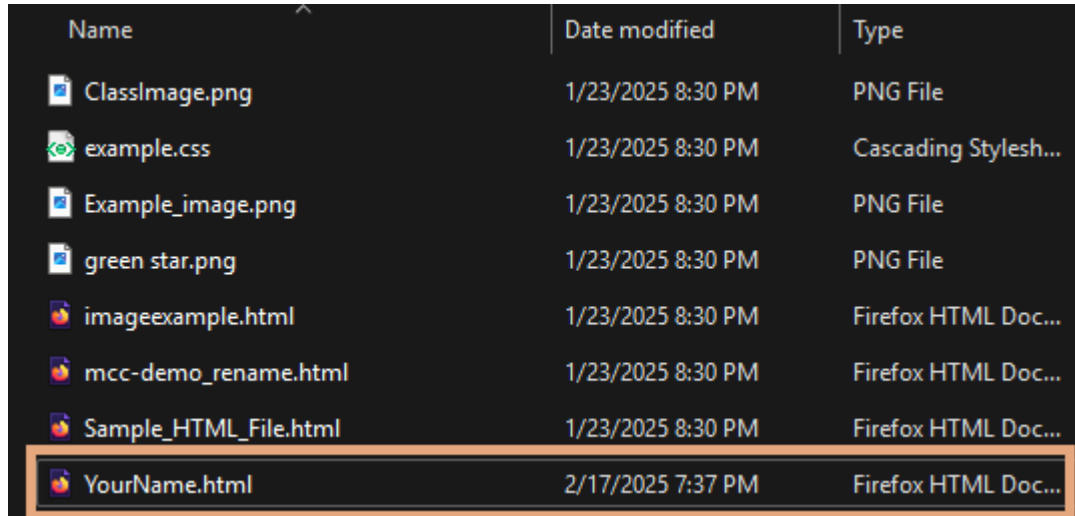
Name	Date modified	Type
 ClassImage	1/23/2025 8:30 PM	PNG File
 example	1/23/2025 8:30 PM	Cascading Stylesh...
 Example_image	1/23/2025 8:30 PM	PNG File
 green star	1/23/2025 8:30 PM	PNG File
 imageexample	1/23/2025 8:30 PM	Firefox HTML Doc...
 mcc-demo_rename	1/23/2025 8:30 PM	Firefox HTML Doc...
 Sample_HTML_File	1/23/2025 8:30 PM	Firefox HTML Doc...
 YourName.html	2/17/2025 7:20 PM	TXT File

You can tell because you can see the `.html`, but the other HTML files (`imageexample`, `mcc-demo_rename`, and `Sample_HTML_File`) have the extension hidden. The icon is also for NotePad++, which is what I have text files mapped to, instead of the browser icon (my default is FireFox, you may have Chrome or Edge). Also, the Type is listed as **TXT File**.

To fix this, follow the instructions in *File Explorer configuration* in the Tips and Tricks handout (in the same directory as this PDF). You should then see that the file is actually called `YourName.html.txt`. Rename the file, removing the trailing `.txt`.

6. Click **Yes** to change the file extension.

The file should be named *YourName.html*.



Name	Date modified	Type
 ClassImage.png	1/23/2025 8:30 PM	PNG File
 example.css	1/23/2025 8:30 PM	Cascading Stylesh...
 Example_image.png	1/23/2025 8:30 PM	PNG File
 green_star.png	1/23/2025 8:30 PM	PNG File
 imageexample.html	1/23/2025 8:30 PM	Firefox HTML Doc...
 mcc-demo_rename.html	1/23/2025 8:30 PM	Firefox HTML Doc...
 Sample_HTML_File.html	1/23/2025 8:30 PM	Firefox HTML Doc...
 YourName.html	2/17/2025 7:37 PM	Firefox HTML Doc...

Congratulations! You have made a blank file with the HTML extension.

Open the HTML file in Notepad++

There are many HTML editors out there, but we're going to use Notepad++.

In File Explorer, right-click *YourName.html* and select **Edit with Notepad++**.

If you have Windows 11, you may have to select **Show more options** before you can select **Edit with Notepad++**.

The blank file, *YourName.html* opens in NotePad++.

You are now ready to add your first heading or title.

Set up the HTML file

HTML has some text that's required.

Your blank file should already be open in Notepad++. If it isn't, see [Open the HTML file in Notepad++](#) on page 41.

1. A properly formatted HTML5 file requires a doctype declaration. The first line of the file must be the following:

```
<!DOCTYPE html>
```

2. After the doctype declaration, start the HTML file using an `<html>` tag.

```
<!DOCTYPE html>
<html>

</html>
```

HTML5 can be lazy or non-strict, meaning that you don't have to close every tag that you open. However, if you don't, the browsers make guesses, which may or may not be what you intended. If you always open and close your tags, it can make debugging easier. I strongly recommend you open (`<html>`) and close (`</html>`) each tag as you type it. It's just a good habit to make your life easier in the future.

3. Add the `<head>` tag to the file, inside of the `<html>` tag.

```
<!DOCTYPE html>
<html>
  <head></head>
</html>
```

The `<head>` tag contains metadata and various technical bits to help make the HTML work.

4. Inside of the `<head>` tag, add the `<title>` tag, with some text.

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
</html>
```

There is a lot more that can go into the `<head>` of an HTML file, but that is for later study.

5. After the `<head>` tag, add a `<body>` tag.

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body></body>
</html>
```

You are now ready to start adding content to your HTML file.

Add a title to the HTML file

You can add content in most any order in an HTML file, but in general, you start with a title or heading.

You need to have your HTML file set up. If it isn't, see [Set up the HTML file](#) on page 41.

- Inside the `<body>` tag, add an `<h1>` tag, with the title text.
For example:

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body>
    <h1>Look a Title</h1>
  </body>
</html>
```

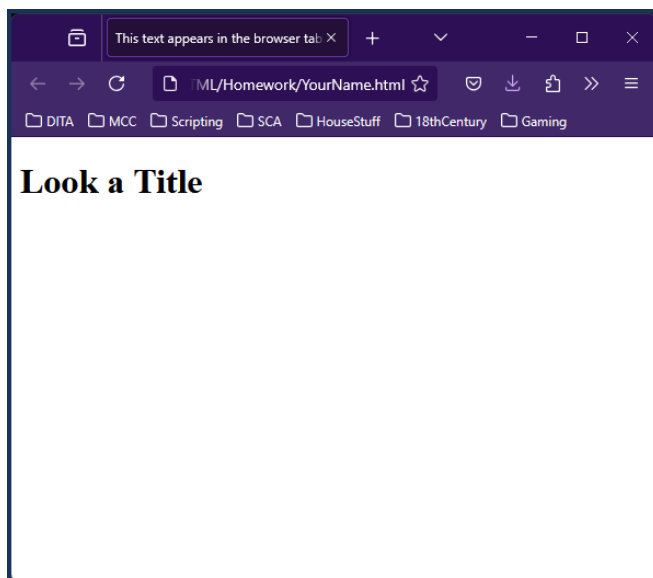
Often, the text of the `<title>` and the first `<h1>` tags are the same. In general, the first `<h1>` tag displays as the title of the page. In many cases, people like the text that displays on the browser tab to match the "title" of the web page. However, this is not always true.

You'll notice in the example that there are spaces or tabs to make the tags inside of other tags appear "nested". This is not required, but it can help make the HTML more readable by humans. Some tools can do this for you, it's often called "Pretty Print" or something similar.

At this point, you can save the file and see your changes in a browser.

1. Save your file in NotePad++.
2. From File Explorer, double-click your HTML file to open it in your default browser.

The HTML file should open in your browser and look something like the following.



Add a paragraph to the HTML file

After the title, insert a paragraph.

1. After the `<h1>` tag, add a `<p>` tag.

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body>
    <h1>Look a Title</h1>
    <p></p>
  </body>
</html>
```

2. Enter the text of your paragraph.

For example, enter the following:

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body>
    <h1>Look a Title</h1>
    <p>The above is a title. It could be formatted as something
specifically
labeled a title, or maybe just a heading level one.</p>
  </body>
</html>
```

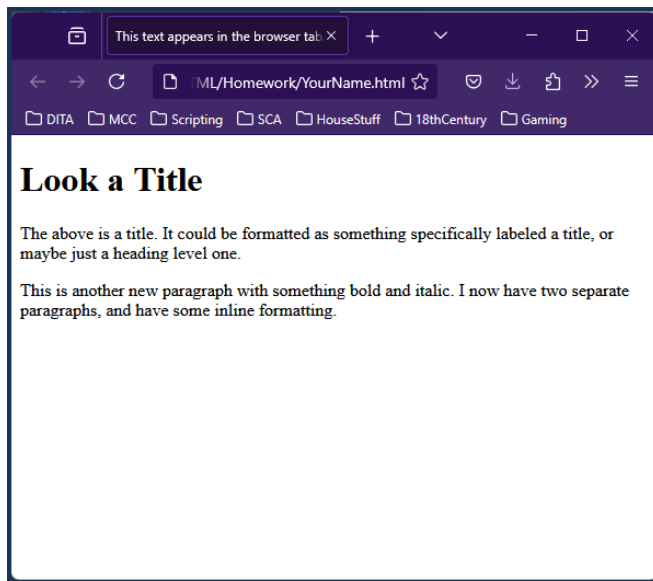
3. Add a second paragraph by adding more content inside another `<p>` tag.

You should have a title with two paragraphs, such as the following:

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body>
    <h1>Look a Title</h1>
    <p>The above is a title. It could be formatted as something
specifically
labeled a title, or maybe just a heading level one.</p>
    <p>This is another new paragraph with something bold and
italic. I now
have two separate paragraphs, and have some inline formatting.</p>
  </body>
</html>
```

If you would like, you can save your file in NotePad++ and look at it in your browser. If you already have the HTML file open in your browser, you can just refresh your browser. If you need instruction for how to view the file in your browser, see the end of [Add a title to the HTML file](#) on page 43.

Your file should look similar to the following:



Add inline formatting to the HTML file

Inline formatting in HTML is done with additional tagging.

You have a paragraph in your HTML file.

HTML works with another language called Cascading Style Sheets (CSS) to provide information about how the HTML tags should display. This CSS can make most any element bold or italic or green or underlined or almost any other style you can think of. Be aware that the CSS of the HTML file you are working with may change the display of any tag you use.

Right now, with the file we are creating, there's no CSS configured. Therefore, we can use the default tags.

To make text italic, you wrap it with an `<em>` (emphasis) tag.

To make text bold, you wrap it with an `<strong>` tag.

Note: In the following procedures, only the relevant sections of the HTML file will be shown. You can see the full file in [Example HTML file](#) on page 54.

1. Place your cursor before the text you want to make italic and type `<em>`. Here is the second paragraph with the `<em>` tag opened.

```
<p>This is another new paragraph with something bold and <em>italic.  
I now have  
two separate paragraphs, and have some inline formatting.</p>
```

2. Place your cursor at the end of the text you want to make italic and close the `<em>` tag.

Here is the second paragraph with the `<em>` tag closed.

```
<p>This is another new paragraph with something bold and <em>italic</em>. I now  
have two separate paragraphs, and have some inline formatting.</p>
```

3. Place your cursor before the text you want to make bold and open the `<strong>` tag.
Here is the second paragraph with the start of the `<strong>` tag.

```
<p>This is another new paragraph with something <strong>bold and  
<em>italic</em>.  
I now have two separate paragraphs, and have some inline  
formatting.</p>
```

4. Place your cursor at the end of the text you want to make bold and close the `<strong>` tag.
Here is the second paragraph with the end `<strong>` tag added.

```
<p>This is another new paragraph with something <strong>bold</strong>  
and  
<em>italic</em>. I now have two separate paragraphs, and have some  
inline formatting.</p>
```

You now have inline formatting marked. You are ready to make an unordered list.

Add an unordered list to the HTML file

Unordered lists are sometimes known as bulleted lists.

The tag that defines an unordered list is the `<ul>` (unordered list) tag. Each bullet in the list is identified as an `<li>` tag.

1. Make sure your cursor is where you want to insert the unordered list.

Unordered lists must be inside the `<body>` tag to appear on the HTML page. (There are lots of other places inside of the `<body>` tag the list could go, such as inside a table or inside another list.)

If you're following along with the example, have your cursor after the second `<p>` tag.

2. Add an `<ul>` tag.

```
<p>This is another new paragraph with something <strong>bold</strong>  
and  
<em>italic</em>. I now have two separate paragraphs, and have some  
inline formatting.</p>  
<ul>  
</ul>
```

3. Inside the `<ul>` tag, add an `<li>` tag with your bullet text inside of it.

```
<p>This is another new paragraph with something <strong>bold</strong>
and
<em>italic</em>. I now have two separate paragraphs, and have some
  inline formatting.</p>
<ul>
  <li>I also need a list.</li>
</ul>
```

4. Add additional `<li>` tags for each bullet.

Your list should resemble the following:

```
<p>This is another new paragraph with something <strong>bold</strong>
and
<em>italic</em>. I now have two separate paragraphs, and have some
  inline formatting.</p>
<ul>
  <li>I also need a list.</li>
  <li>This is a second list item.</li>
  <li>And look, a third list item.</li>
</ul>
```

Add an ordered list to the HTML file

Unordered lists are sometimes known as numbered lists.

The tag that defines an unordered list is the `<ol>` (ordered list) tag. Each bullet in the list is identified as an `<li>` tag.

1. Make sure your cursor is where you want to insert the ordered list.

Ordered lists must be inside the `<body>` tag to appear on the HTML page. (There are lots of other places inside of the `<body>` tag the list could go, such as inside a table or inside another list.)

If you're following along with the example, have your cursor after the `<ul>` tag.

2. Add an `<ol>` tag.

```
</ul>
<ol>
</ol>
```

3. Inside the `<ol>` tag, add an `<li>` tag with your bullet text inside of it.

```
</ul>
<ol>
  <li>Now I need an ordered list.</li>
</ol>
```

4. Add additional `<li>` tags for each bullet.

Your list should resemble the following:

```
<ol>
  <li>Now I need an ordered list.</li>
  <li>I can use this for procedures.</li>
  <li>Or to identify items in an image I don't want to translate.</li>
</ol>
```

Insert an external link in the HTML file

You often need to reference external websites when writing.

These instructions are how to link to an external web site. Links are made using the <a> tag.

Links (the <a> tag) need to be inside of a "block" tag. They are a type of inline tag.

1. Add a paragraph where you want to insert your link.

If you're following along, add the <p> tag after the tag.

```
</ol>
<p>This is a link </p>
```

This paragraph also includes some introductory text for the link.

2. Start the <a> tag.

```
</ol>
<p>This is a link <a> </p>
```

3. Add an @href attribute to the <a> tag and set the attribute value to the URL.

```
</ol>
<p>This is a link <a href="https://www.google.com"> </p>
```

4. Add the text for your link and close the <a> tag.

For example:

```
</ol>
<p>This is a link <a href="https://www.google.com">to Google</a>.</p>
```


Insert an internal link in the HTML file

Internal links, meaning to some paragraph or heading on the same page are also useful.

On a long HTML page, you often have links to section headings. You may also have periodic "link to top" links to easily get back to the top of the page.

Internal links are also made using the <a> tag, but you also have to identify the target for the link. In HTML5, the target is identified by adding an @id attribute to the tag you want the link to go to.

1. Add an @id attribute to the first <h1> in the file.

For example:

```
<body>
  <h1 id="myLinkTarget">Look a Title</h1>
```

2. Add a paragraph where you want to insert your link.

You don't have to make a new paragraph, you could add the link to an existing paragraph. If you're following along, add the <p> tag after the last <p> tag.

```
</ol>
<p>This is a link <a href="https://www.google.com">to Google</a>.</p>
<p>This is a link </p>
```

This paragraph also includes some introductory text for the link.

3. Start the <a> tag.

```
</ol>
<p>This is a link <a href="https://www.google.com">to Google</a>.</p>
<p>This is a link <a></p>
```

This paragraph also includes some introductory text for the link.

4. Add an @href attribute to the <a> tag and set the attribute value to the @id attribute you want to link to, preceded by a #.

```
</ol>
<p>This is a link <a href="https://www.google.com">to Google</a>.</p>
<p>This is a link <a href="#myLinkTarget"></p>
```

5. Add the text for your link and close the <a> tag.

For example:

```
</ol>
<p>This is a link <a href="https://www.google.com">to Google</a>.</p>
<p>This is a link <a href="#myLinkTarget">to the first heading</a>.</p>
```

Insert an image in the HTML file

Images in HTML are inserted by reference.

You need to have a web-friendly image in the folder you made in [Make an HTML file](#) on page 40. For example, you can reference a JPG, GIF, or PNG file.

You can also use a relative path from the HTML file to where the image is. If you have already added an image for another homework assignment, you could reference it. However, that requires that you understand how relative paths work. If you're not ready for that yet, just make sure your image is in your `mcc_tools_tech\Week05-AgileHTML\Homework\YourNameHW` folder.

The example image used is called `graphic.png`.

1. Have your cursor where you want to insert your image.

Images can be added as a block or inline. In general, if it is a large image, you want to treat it as a block. If it is a small image (such as for an icon), you can have it inline.

2. Insert an `<img>` tag.

```
<p>This is a link <a href="#myLinkTarget">to the first heading</a>.</p>
<img/>
```

The `<img>` tag generally does not have any other content inside of it. Therefore, you can open and close the tag in the same set of angle brackets.

3. Add a `<src>` attribute to the `<img>` tag that contains the path to the image.

```
<p>This is a link <a href="#myLinkTarget">to the first heading</a>.</p>

```

4. Technically in HTML5 you should always add an `@alt` attribute in of the `<img>` element to provide alternative text for accessibility.

For example:

```
<p>This is a link <a href="#myLinkTarget">to the first heading</a>.</p>

```

Add a table to the HTML file

Tables are very useful for displaying data, but can be complicated.

A table in HTML is contained within the `<table>` tag.

There are many different ways to create a table that enable you to build very complex tables with merged cells, different sections of the table, etc. This example only shows the bare minimum required to make a table.

Tables are constructed of rows. You identify rows with the `<tr>` (table row) tag.

(Yes, tables have columns, but without special scripting, you can't select a column in a table, so it's generally better to contemplate HTML tables as rows.)

Inside of each row are the *cells* of the table. You can use `<th>` (table heading) tag for any cell you want to be a header or label for the row or column. Regular cells are in `<td>` (table data) tags. Unless you do something interesting (not covered in this course), every row in your table should have the same number of cells.

In this example we're making a two column table with a header row and three body rows.

1. Insert a `<table>` tag.

```
<table>
</table>
```

2. Inside of the `<table>` tag, add a `<tr>` tag.

```
<table>
  <tr></tr>
</table>
```

3. Inside of the `<tr>` tag, add two `<th>` tags and add the heading text.

```
<table>
  <tr>
    <th>Table Head 1</th>
    <th>Table Head Column 2</th>
  </tr>
</table>
```

4. After the first row, add another three <tr> tags.

```
<table>
  <tr>
    <th>Table Head 1</th>
    <th>Table Head Column 2</th>
  </tr>
  <tr>
  </tr>
  <tr>
  </tr>
  <tr>
  </tr>
</table>
```

5. Inside each of these <tr> tags, add two <td> tags.

```
<table>
  <tr>
    <th>Table Head 1</th>
    <th>Table Head Column 2</th>
  </tr>
  <tr>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
  </tr>
  <tr>
    <td></td>
    <td></td>
  </tr>
</table>
```

6. Add content to your table cells.

For example, here is a completed table:

```
<table>
  <tr>
    <th>Table Head 1</th>
    <th>Table Head Column 2</th>
  </tr>
  <tr>
    <td>Column 1</td>
    <td>Column 2</td>
  </tr>
  <tr>
    <td>More Column 1</td>
    <td>More Column 2</td>
  </tr>
  <tr>
    <td>And a third row for fun</td>
    <td>With one more column 2</td>
  </tr>
</table>
```

Example HTML file

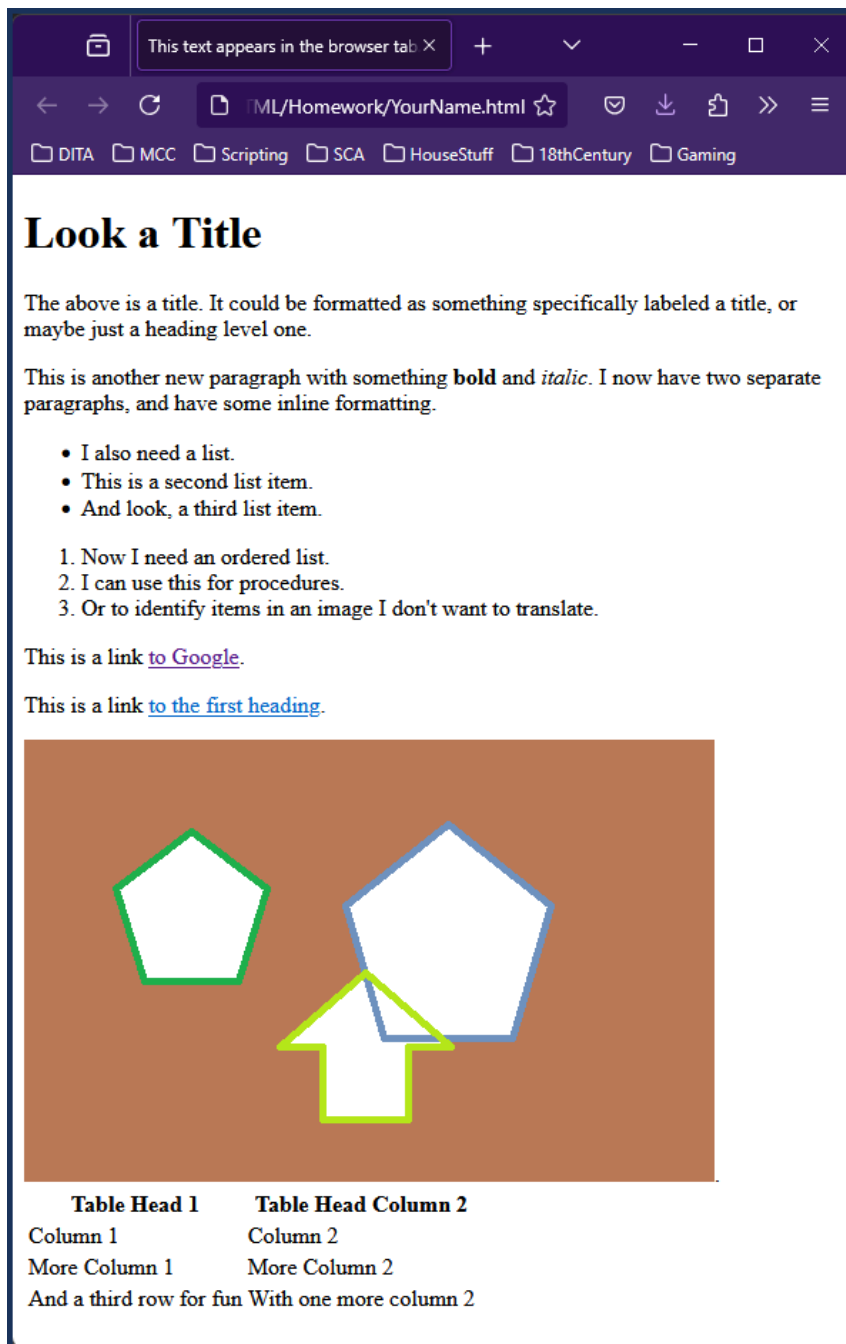
If you followed the previous instructions, your file should resemble the following.

```
<!DOCTYPE html>
<html>
  <head><title>This text appears in the browser tab.</title></head>
  <body>
    <h1 id="myLinkTarget">Look a Title</h1>
    <p>The above is a title. It could be formatted as something
specifically
labeled a title, or maybe just a heading level one.</p>
    <p>This is another new paragraph with something <strong>bold</
strong> and
<em>italic</em>. I now have two separate paragraphs, and have some
inline formatting.</p>
    <ul>
      <li>I also need a list.</li>
      <li>This is a second list item.</li>
      <li>And look, a third list item.</li>
    </ul>
    <ol>
      <li>Now I need an ordered list.</li>
      <li>I can use this for procedures.</li>
      <li>Or to identify items in an image I don't want to
translate.</li>
    </ol>
    <p>This is a link <a href="https://www.google.com">to Google</
a>.</p>
    <p>This is a link <a href="#myLinkTarget">to the first
heading</a>.</p>
    .
    <table>
      <tr>
        <th>Table Head 1</th>
        <th>Table Head Column 2</th>
      </tr>
      <tr>
        <td>Column 1</td>
        <td>Column 2</td>
      </tr>
      <tr>
        <td>More Column 1</td>
        <td>More Column 2</td>
      </tr>
      <tr>
        <td>And a third row for fun</td>
        <td>With one more column 2</td>
      </tr>
    </table>
  </body>
</html>
```

Save your file and upload it to GitHub. If you need a refresher on working with GitHub, see Handouts/git_cheatsheet.pdf.

Congratulations! You've made an HTML file!

It should look similar to the following:



This file and the image are available in the Week05-AgileHTML\Homework folder.

Week 6 GitHub Flavored Markdown

There are multiple flavors of markdown out there in the world. We are going to focus on GitHub Flavored Markdown.

These instructions should help you complete the minimum requirements for the homework.

These instructions should be used in conjunction with the information in the following files:

- [mcc_tools_tech\Week06-LightweightMarkup\markdown.pdf](#)
- [mcc_tools_tech\Week06-LightweightMarkup\week6_Lightweight_markup.html](#)

In this section:

- [Make a md file on page 57](#)
Markdown is a text file.
- [Open the md file in Notepad++ on page 57](#)
There are a few markdown editors out there, but we're going to use Notepad++.
- [Add a title to the markdown file on page 57](#)
The first block you need to add is a title or heading.
- [Add a paragraph to the markdown file on page 58](#)
After the title, insert a paragraph.
- [Add inline formatting to the markdown file on page 58](#)
Inline formatting in GitHub Flavored Markdown is done with underscores and asterisks.
- [Add an unordered list to the markdown file on page 59](#)
Unordered lists are sometimes known as bulleted lists.
- [Add an ordered list to the markdown file on page 60](#)
Unordered lists are sometimes known as numbered lists.
- [Insert a link in the markdown file on page 61](#)
You often need to reference external websites when writing.
- [Insert an image in the markdown file on page 61](#)
Images in markdown are inserted by reference.
- [Add a table to the markdown file on page 62](#)
Tables are very useful for displaying data but complicated in markdown.
- [Example markdown file on page 63](#)
If you followed the previous instructions, your file should resemble the following.

Make a md file

Markdown is a text file.

1. In File Explorer, right-click in the `mcc_tools_tech\Week06-LightweightMarkup\Homework` folder and select **New > Folder**.
This makes a new folder in the Homework directory. It should be called `New Folder`, and the text "New Folder" should be highlighted and ready to rename the folder.
2. Rename the folder *YourNameHW*.
For example, I would name the folder `ZoeLawsonHW`.
I will keep referring to the folder as *YourNameHW*.
3. Navigate into the *YourNameHW* folder.
4. Still in File Explorer, right-click in *YourNameHW* and select **New > Text Document**.
This makes a new text document in the *YourNameHW* folder. It should be called `New Text Document .txt`, and the text "New Text Document" should be highlighted and ready to rename the file.
5. Rename the text file *YourName .md*.
For example, I would name the file `ZoeLawson .md`.
You must change the extension to `md`.
You should receive a message stating If you change a file name extension, the file might become unusable. Are you sure you want to change it?.
6. Click **Yes** to change the file extension.
The file should be named *YourName .md*.

Congratulations! You have made a markdown file.

Open the md file in Notepad++

There are a few markdown editors out there, but we're going to use Notepad++.

In File Explorer, right-click *YourName .md* and select **Edit with Notepad++**.

If you have Windows 11, you may have to select **Show more options** before you can select **Edit with Notepad++**.

The blank file, *YourName .md* opens in NotePad++.

You are now ready to add your first heading or title.

Add a title to the markdown file

The first block you need to add is a title or heading.

Your blank file should already be open in Notepad++. If it isn't, see [Open the md file in Notepad++](#) on page 57.

1. On the first line, type a `#` followed by a space, followed by your title text.

For example:

```
# Look a Title
```

2. After entering the text of the title, make sure there is a blank line after the title.

Your file should look like the following:

```
# Look a Title
```

Notice that there is a blank line after the title, and your cursor is at the start of line 3.

Add a paragraph to the markdown file

After the title, insert a paragraph.

Your cursor should be at the beginning of line 3.

1. Enter the text of your paragraph.

For example, enter the following:

```
The above is a title. It could be formatted as something specifically  
labeled a title, or maybe just a heading level one.
```

2. After entering the text of the paragraph, make sure there is a blank line after the paragraph.
3. Add a second paragraph.

You should have a title with two paragraphs, such as the following:

```
# Look a Title
```

```
The above is a title. It could be formatted as something specifically  
labeled a title, or maybe just a heading level one.
```

```
This is another new paragraph with something bold and italic. I now  
have two separate paragraphs, and have some inline formatting.
```

Notice that there is an empty line between every block of text. Your cursor should be at the beginning of line 7.

Add inline formatting to the markdown file

Inline formatting in GitHub Flavored Markdown is done with underscores and asterisks.

You have a paragraph in your markdown file.

To make text italic, you wrap it with a single underscore or asterisk.

To make text bold, you wrap it in two underscores or two asterisks.

1. Place your cursor before the text you want to make italic and insert a single underscore or a single asterisk.

Here is the second paragraph with the first underscore added.

```
This is another new paragraph with something bold and _italic_. I now  
have two separate paragraphs, and have some inline formatting.
```

2. Place your cursor at the end of the text you want to make italic and insert a single underscore or a single asterisk.

You have to use the same symbol as you used to start the italic text. `_italic_` or `*italic*` work; `_italic*` or `*italic_` does not work.

Here is the second paragraph with the second underscore added.

```
This is another new paragraph with something bold and _italic_. I now  
have two separate paragraphs, and have some inline formatting.
```

3. Place your cursor before the text you want to make bold and insert two underscores or two asterisks.

Here is the second paragraph with the first set of asterisks added.

```
This is another new paragraph with something **bold and _italic_. I  
now have two separate paragraphs, and have some inline formatting.
```

4. Place your cursor at the end of the text you want to make bold and insert two underscores or two asterisks.

You have to use the same symbol as you used to start the italic text. `__bold__` or `**bold**` work; `__bold**` or `**bold__` does not work.

Here is the second paragraph with the second set of asterisks added.

```
This is another new paragraph with something **bold** and _italic_. I  
now have two separate paragraphs, and have some inline formatting.
```

You now have inline formatting marked. You are ready to make an unordered list.

Add an unordered list to the markdown file

Unordered lists are sometimes known as bulleted lists.

Make sure your cursor is at the start of a new line and that there is a blank line above your cursor.

You can use many different characters to indicate an unordered list. The most common are asterisk or hyphens.

1. Have your cursor at the start of the line, and confirm there is a blank line between where your cursor is and the previous content.

If you are following the example in these instructions, your cursor should be at the start of line 7.

2. Insert a hyphen, a space, and then the text of your list item.

For example:

```
- I also need a list.
```

3. Press Enter.
4. Insert a hyphen, a space, and then the text of your second list item.
You do not need to have blank lines between list items.
5. Repeat entering a hyphen, a space, and then the text of your list item.
6. Make sure there is a blank line after your list.

Your list should resemble the following:

```
- I also need a list.  
- This is a second list item.  
- And look, a third list item.
```

Add an ordered list to the markdown file

Unordered lists are sometimes known as numbered lists.

You make a numbered list item by entering the number 1, a period, a space, and then the text of the list item.

You should only use the number 1, which can be confusing. Some editors may recognize if you actually enter the numbers, but it's not guaranteed.

1. Have your cursor at the start of the line, and confirm there is a blank line between where your cursor is and the previous content.
If you are following the example in these instructions, your cursor should be at the start of line 11.
2. Insert the number 1, a period, a space, and then the text of your list item.
For example:

```
1. Now I need an ordered list.
```

3. Press Enter.
4. Insert a number 1, a period, a space, and then the text of your second list item.
You do not need to have blank lines between list items.
5. Repeat entering a number 1, a period, a space, and then the text of your list item.
6. Make sure there is a blank line after your list.

Your list should resemble the following:

```
1. Now I need an ordered list.  
1. I can use this for procedures.  
1. Or to identify items in an image I don't want to translate.
```

Insert a link in the markdown file

You often need to reference external websites when writing.

These instructions are how to link to an external web site. Linking to internal locations require different formatting, and is not required for this homework.

If you're interested in internal links, see <https://stackoverflow.com/questions/27981247/github-markdown-same-page-link>.

1. Have your cursor where you want to insert your link.
2. Type an open square bracket [.
3. Enter the text for the link.
4. Type a close square bracket].
5. Type an open parenthesis (.
6. Enter the URL for the link.

You have to enter the full URL, including the `http://`.

7. Type a close parenthesis).

For example:

```
This is a [link to Google](http://www.google.com).
```

Insert an image in the markdown file

Images in markdown are inserted by reference.

You need to have a web-friendly image in the folder you made in [Make a md file](#) on page 57. For example, you can reference a JPG, GIF, or PNG file.

You can also use a relative path from the markdown file to where the image is. If you have already added an image for another homework assignment, you could reference it. However, that requires that you understand how relative paths work. If you're not ready for that yet, just make sure your image is in your `mcc_tools_tech\Week06-LightweightMarkup\Homework\YourNameHW` folder.

The example image used is called `graphic.png`.

1. Have your cursor where you want to insert your image.
2. Type an exclamation point ! followed by an open square bracket [.
3. Enter your alternative text for the image.
This is used by screen readers or if your image cannot be found.
4. Type a close square bracket].
5. Type an open parenthesis {.
6. Enter the relative path to your image.

If your image is in the same folder as your markdown file, it should just be the file name.

7. Type a close parenthesis).
8. If this is a standalone image, make sure there is a blank line before and after the line of text for the image.

For example:

```
![A few overlapping geometric shapes in different colors.](graphic.png)
```

Add a table to the markdown file

Tables are very useful for displaying data but complicated in markdown.

Tables are defined using pipes | and hyphens -.

1. Have your cursor at the start of the line, and confirm there is a blank line between where your cursor is and the previous content.
2. Type a pipe |, a space, the text of your first column header, another space, and then another pipe |.
3. Type a space, the text of your second column header, another space, and then another pipe |.
4. Repeat step 3 for every column.
5. Press **Enter** at the end of your table row.
6. For the second line to mark the previous line as the table head, type a pipe | a space, then a number of hyphens - equal to the text of your first column, another space, and then another pipe |. Repeat the hyphens and pipes to match your table head.

For example:

Table Head 1	Table Head Column 2
-----	-----

7. For each row of your table, type a pipe |, a space, the text of your cell, a space, a pipe, the text of the second column cell, a space, and a pipe. You want to end your row with a space and a pipe.

This means that most likely your pipes will not line up because your cell text will be different lengths.

8. Make sure there is a blank line after your table.

For example, here is the completed table:

```
| Table Head 1 | Table Head Column 2 |
| ----- | ----- |
| Column 1 | Column 2 |
| More Column 1 | More Column 2 |
| And a third row for fun | With one more column 2 |
```

Example markdown file

If you followed the previous instructions, your file should resemble the following.

```
# Look a Title
```

The above is a title. It could be formatted as something specifically labeled a title, or maybe just a heading level one.

This is another new paragraph with something **bold** and *italic*. I now have two separate paragraphs, and have some inline formatting.

- I also need a list.
- This is a second list item.
- And look, a third list item.

1. Now I need an ordered list.
1. I can use this for procedures.
1. Or to identify items in an image I don't want to translate.

This is a [link to Google](http://www.google.com).

![A few overlapping geometric shapes in different colors.](graphic.png)

```
| Table Head 1 | Table Head Column 2 |
| ----- | ----- |
| Column 1 | Column 2 |
| More Column 1 | More Column 2 |
| And a third row for fun | With one more column 2 |
```

Save your file and upload it to GitHub. If you need a refresher on working with GitHub, see [Handouts/git_cheatsheet.pdf](#).

Congratulations! You've made a GitHub Flavored Markdown file!

Week 7 Restructured Text

ReStructured Text (ReST) is another type of lightweight markup.

These instructions should help you meet the minimum requirements for the assignment.

These instructions should be used in conjunction with the information in the following files:

- [mcc_tools_tech\Week07-MoreLightweightMarkup\reStructuredText.pdf](#)
- [mcc_tools_tech\Week07-MoreLightweightMarkup\week7_more_lightweight_markup.html](#)

See also [Quick reStructured Text](#) for a brief description of the syntax.

In this section:

- [Make a rst file on page 65](#)
ReStructured Text is a text file.
- [Open the rst file in Notepad++ on page 65](#)
ReStructured Text was created to be used with applications written in Python. A Python environment editor should handle ReST, but we're going to use Notepad++.
- [Add a title to the markdown file on page 65](#)
The first block you need to add is a title or heading.
- [Add a paragraph to the ReST file on page 66](#)
After the title, insert a paragraph.
- [Add inline formatting to the ReST file on page 66](#)
Inline formatting in ReST is done with underscores and asterisks.
- [Add an unordered list to the ReST file on page 67](#)
Unordered lists are sometimes known as bulleted lists.
- [Add an ordered list to the ReST file on page 68](#)
Unordered lists are sometimes known as numbered lists.
- [Insert a link in the ReST file on page 68](#)
You often need to reference external websites when writing.
- [Insert an image to the ReST file on page 69](#)
Images in ReST are inserted by reference.
- [Add a table to the ReST file on page 70](#)
Tables are very useful for displaying data but complicated in ReST.
- [Example ReST file on page 71](#)
If you followed the previous instructions, your file should resemble the following.

Make a rst file

ReStructured Text is a text file.

1. In File Explorer, right-click in the `mcc_tools_tech\Week07-MoreLightweightMarkup\Homework` folder and select **New > Folder**.
This makes a new folder in the Homework directory. It should be called `New Folder`, and the text "New Folder" should be highlighted and ready to rename the folder.
2. Rename the folder `YourNameHW`.
For example, I would name the folder `ZoeLawsonHW`.
I will keep referring to the folder as `YourNameHW`.
3. Navigate into the `YourNameHW` folder.
4. Still in File Explorer, right-click in `YourNameHW` and select **New > Text Document**.
This makes a new text document in the `YourNameHW` folder. It should be called `New Text Document .txt`, and the text "New Text Document" should be highlighted and ready to rename the file.
5. Rename the text file `YourName.rst`.
For example, I would name the file `ZoeLawson.rst`.
You must change the extension to `.rst`.
You should receive a message stating If you change a file name extension, the file might become unusable. Are you sure you want to change it?.
6. Click **Yes** to change the file extension.
The file should be named `YourName.rst`.

Congratulations! You have made a ReStructured text file.

Open the rst file in Notepad++

ReStructured Text was created to be used with applications written in Python. A Python environment editor should handle ReST, but we're going to use Notepad++.

In File Explorer, right-click `YourName.rst` and select **Edit with Notepad++**.

If you have Windows 11, you may have to select **Show more options** before you can select **Edit with Notepad++**.

The blank file, `YourName.rst` opens in Notepad++.

You are now ready to add your first heading or title.

Add a title to the markdown file

The first block you need to add is a title or heading.

Your blank file should already be open in Notepad++. If it isn't, see [Open the rst file in Notepad++ on page 65](#).

1. On the first line, type your title text and press **Enter**.

2. On the second line, type an equal sign = the same length as your title.
You can actually use one of many characters, including +, #, *, or ~.
3. After entering the text of the title, and the line of characters marking the title, make sure there is a blank line after the title.

Your file should look like the following:

```
Look a Title
=====
```

Notice that there is a blank line after the title, and your cursor is at the start of line 4.

Add a paragraph to the ReST file

After the title, insert a paragraph.

Your cursor should be at the beginning of line 4.

1. Enter the text of your paragraph.
For example, enter the following:

```
The above is a title. It could be formatted as something specifically
labeled a title, or maybe just a heading level one.
```

2. After entering the text of the paragraph, make sure there is a blank line after the paragraph.
3. Add a second paragraph.

You should have a title with two paragraphs, such as the following:

```
Look a Title
=====

The above is a title. It could be formatted as something specifically
labeled a title, or maybe just a heading level one.

This is another new paragraph with something bold and italic. I now
have two separate paragraphs, and have some inline formatting.
```

Notice that there is an empty line between every block of text. Your cursor should be at the beginning of line 8.

Add inline formatting to the ReST file

Inline formatting in ReST is done with underscores and asterisks.

You have a paragraph in your ReST file.

To make text italic, you wrap it with a single asterisk.

To make text bold, you wrap it in two asterisks.

1. Place your cursor before the text you want to make italic and insert a single asterisk.
Here is the second paragraph with the first asterisk added.

```
This is another new paragraph with something bold and *italic. I now  
have two separate paragraphs, and have some inline formatting.
```

2. Place your cursor at the end of the text you want to make italic and insert a single asterisk.
Here is the second paragraph with the second underscore added.

```
This is another new paragraph with something bold and *italic*. I now  
have two separate paragraphs, and have some inline formatting.
```

3. Place your cursor before the text you want to make bold and insert two asterisks.
Here is the second paragraph with the first set of asterisks added.

```
This is another new paragraph with something **bold and *italic*. I  
now have two separate paragraphs, and have some inline formatting.
```

4. Place your cursor at the end of the text you want to make bold and insert two asterisks.
Here is the second paragraph with the second set of asterisks added.

```
This is another new paragraph with something **bold** and *italic*. I  
now have two separate paragraphs, and have some inline formatting.
```

You now have inline formatting marked. You are ready to make an unordered list.

Add an unordered list to the ReST file

Unordered lists are sometimes known as bulleted lists.

Make sure your cursor is at the start of a new line and that there is a blank line above your cursor.

You can use many different characters to indicate an unordered list. The most common are asterisk or hyphens.

1. Have your cursor at the start of the line, and confirm there is a blank line between where your cursor is and the previous content.
If you are following the example in these instructions, your cursor should be at the start of line 8.
2. Insert a hyphen, a space, and then the text of your list item.
For example:

```
- I also need a list.
```

You can use a hyphen -, an asterisk *, or a plus sign + to mark an unordered list.

3. Press Enter.
4. Insert a hyphen, a space, and then the text of your second list item.
You do not need to have blank lines between list items.
5. Repeat entering a hyphen, a space, and then the text of your list item.

6. Make sure there is a blank line after your list.

Your list should resemble the following:

```
- I also need a list.  
- This is a second list item.  
- And look, a third list item.
```

Add an ordered list to the ReST file

Unordered lists are sometimes known as numbered lists.

You make a numbered list item by entering a pound sign/hashtag/octothorpe #, a period, a space, and then the text of the list item.

The pound sign makes an autonumbered list. You can use exact numbers if you need a list to start at a number other than 1.

1. Have your cursor at the start of the line, and confirm there is a blank line between where your cursor is and the previous content.

If you are following the example in these instructions, your cursor should be at the start of line 12.

2. Insert a pound sign #, a period, a space, and then the text of your list item.
For example:

```
#. Now I need an ordered list.
```

3. Press Enter.
4. Insert a pound sign #, a period, a space, and then the text of your second list item.
You do not need to have blank lines between list items.
5. Repeat entering a pound sign #, a period, a space, and then the text of your list item.
6. Make sure there is a blank line after your list.

Your list should resemble the following:

```
#. Now I need an ordered list.  
#. I can use this for procedures.  
#. Or to identify items in an image I don't want to translate.
```

Insert a link in the ReST file

You often need to reference external websites when writing.

These instructions are how to link to an external web site. Linking to internal locations require different formatting, and is not required for this homework.

ReST links consist of two items:

- A text for the link, followed by an underscore.

If the text has spaces in it, you must wrap the text in a back tick. On US keyboards, the back tick is the key to the left of the 1. It is usually the same key as the tilde ~.

For example: ``A link to google`_`

- A definition for the link, that begins with two periods, a space, an underscore, and the link text.

For example: `.. _`A link to google`: http://www.google.com`

1. Have your cursor where you want to insert your link.
2. Type the text for your link, followed by an underscore `_`.
3. On a separate line, type two periods, a space, an underscore, the name, and a colon.
4. Enter the URL for the link.

You have to enter the full URL, including the `http://`.

For example:

```
This is a `link to Google`_  
.. _`link to Google`: http://www.google.com
```

Insert an image to the ReST file

Images in ReST are inserted by reference.

You need to have a web-friendly image in the folder you made in [Make a rst file](#) on page 65. For example, you can reference a JPG, GIF, or PNG file.

You can also use a relative path from the markdown file to where the image is. If you have already added an image for another homework assignment, you could reference it. However, that requires that you understand how relative paths work. If you're not ready for that yet, just make sure your image is in your `mcc_tools_tech\Week07-MoreLightweightMarkup\Homework\YourNameHW` folder.

The example image used is called `graphic.png`.

1. Have your cursor where you want to insert your image.
2. Type two periods, a space, the text `image` followed by two colons and a space.
3. Enter the relative path to your image.
If your image is in the same folder as your markdown file, it should just be the file name.
4. As a standalone image, make sure there is a blank line before and after the line of text for the image.

For example:

```
.. image:: graphic.png
```

Add a table to the ReST file

Tables are very useful for displaying data but complicated in ReST.

Tables are defined using pipes |, plus signs +, and hyphens -. Table headings are defined with equal signs =.

- The plus sign + is used to mark corners.
- The hyphen - is used to mark row borders.
- The equal sign = is used to mark the table header.
- The pipe | is used to mark column borders.

All of the text of the table has to be surrounded by the demarcation symbols.

It's probably easiest to type out the text of your table, and then make the demarcation symbols around the text.

- This is difficult to explain in words. Hopefully this example will explain better than numerous confusing steps:

```
+-----+-----+
| Table Head 1 | Table Head Column 2 |
+=====+=====+
| Column 1 | Column 2 |
+-----+-----+
| More Column 1 | More Column 2 |
+-----+-----+
| And a third row for fun | With one more column 2 |
+-----+-----+
```

Example ReST file

If you followed the previous instructions, your file should resemble the following.

```
Look a Title
=====

The above is a title. It could be formatted as something specifically
  labeled a title, or maybe just a heading level one.

This is another new paragraph with something bold and italic. I now
  have two separate paragraphs, and have some inline formatting.

- I also need a list.
- This is a second list item.
- And look, a third list item.

#. Now I need an ordered list.
#. I can use this for procedures.
#. Or to identify items in an image I don't want to translate.

This is a `link to Google`_

.. _`link to Google`: http://www.google.com

.. image:: graphic.png

+-----+-----+
| Table Head 1          | Table Head Column 2  |
+=====+=====+
| Column 1              | Column 2              |
+-----+-----+
| More Column 1         | More Column 2         |
+-----+-----+
| And a third row for fun | With one more column 2 |
+-----+-----+
```

Save your file and upload it to GitHub. If you need a refresher on working with GitHub, see [Handouts/git_cheatsheet.pdf](#).

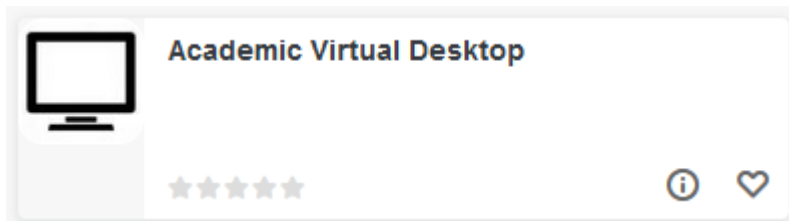
Congratulations! You've made a ReStructured Text file!

Using the Virtual Desktop

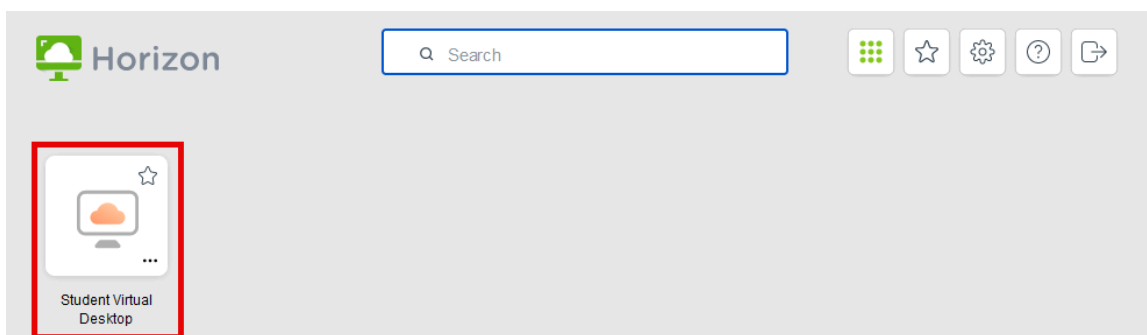
Middlesex Community College provides a virtual student desktop to access applications provided by the school.

The virtual desktop gives you access to all the Microsoft Office applications and Techsmith Camtasia.

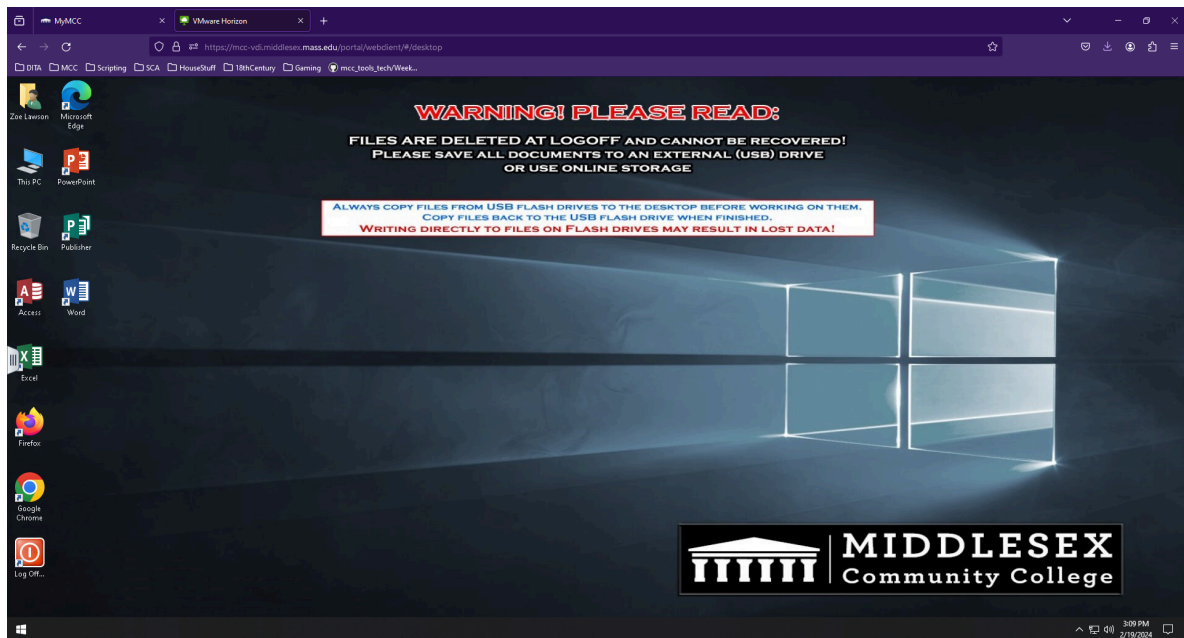
1. Log in to MyMCC (<https://mymcc.middlesex.edu/>).
2. Click **Academic Virtual Desktop**.



3. Sign in to MCC (again).
4. Click **Student Virtual Desktop**.



5. You may be prompted to choose between the desktop client and the browser client. This example uses the browser client.



You now have access to the virtual desktop in your browser window.

Remember that while you might be able to save files locally, there is no guarantee they will still be there the next time you log in. Save any work to your own cloud drive (OneDrive, Google Drive, etc.), or take advantage of the GitHub.com browser tools.